

Fundamentals of Solid State Physics

Electronic Devices

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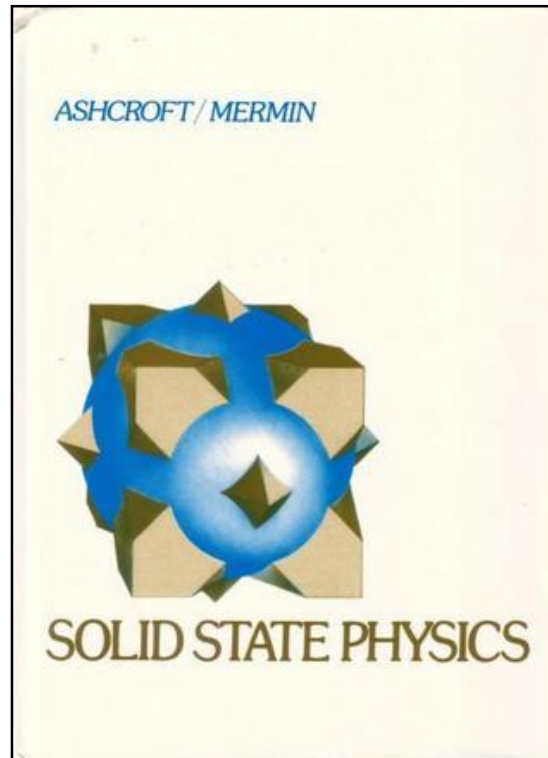


This Class

- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- **Electronic Properties (Week 4–12)**
 - Free electrons (the Drude and Sommerfeld models)
 - Electrons in a periodic potential (Bloch's Theorem)
 - The near-free electron model, the tight-binding model
 - Electronic band diagram, band gaps, effective mass
 - Metals, insulators, semiconductors
 - Devices: junctions, diodes, transistors
- Thermal Properties (Week 13)
- Optical Properties (Week 14)
- Magnetic Properties (Week 15)

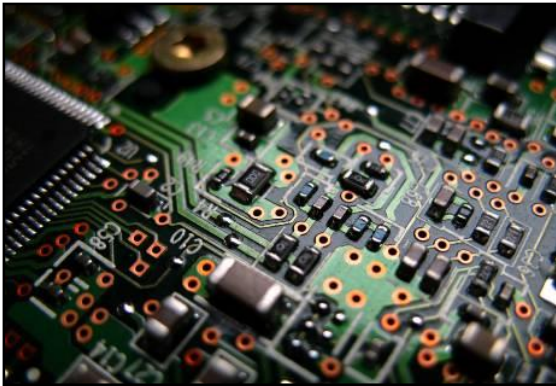
Further Reading

- Ashcroft & Mermin, Chapter 29
- PV Education online course, Chapter 3
 - <https://www.pveducation.org/>



Semiconductors - Applications

semiconductors are the basis of electronics and photonics



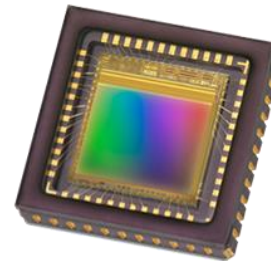
integrated circuits



LEDs



lasers

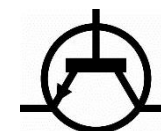


detectors



solar cells

key components: junctions



Junctions

- **Semiconductor-Semiconductor**

- pn homojunction 同质结
- heterojunction 异质结

- **Metal-Metal**

- **Metal-Semiconductor**

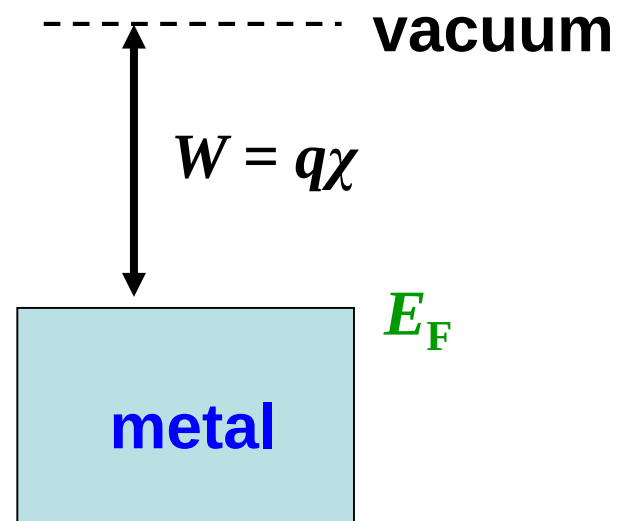
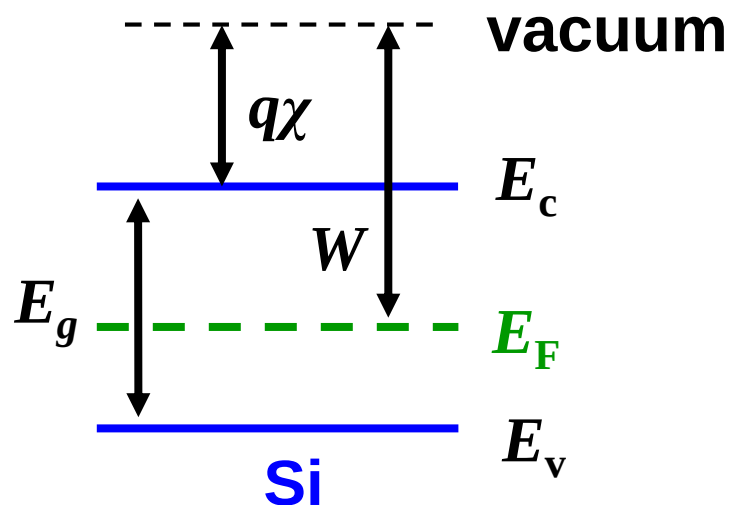
- Ohmic contact
- Schottky contact

- **Metal-Oxide-Semiconductor**

- MOSFET 场效应晶体管

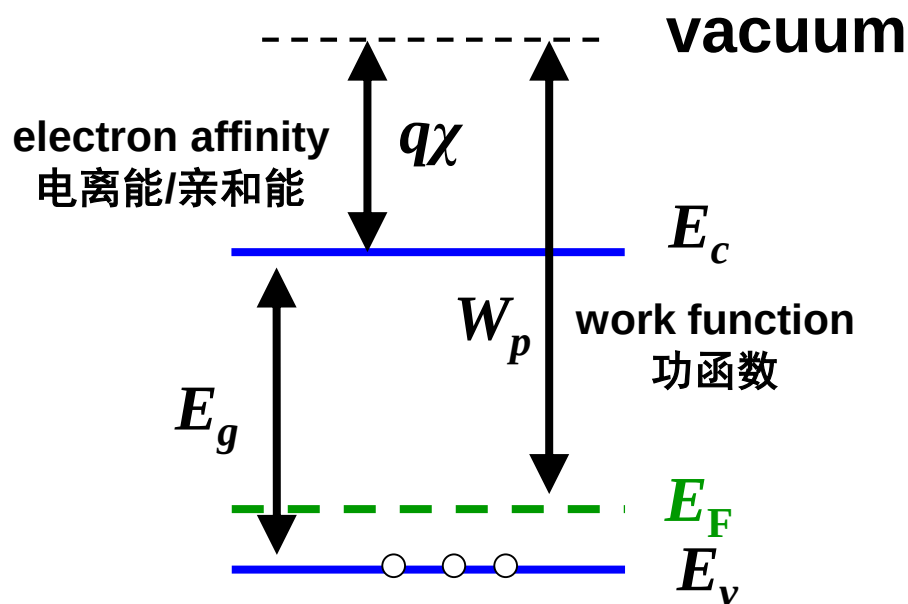
Energy Levels

- Electron affinity (电离能/亲和能) $q\chi$
 - energy difference from conduction band to vacuum
- Work function (功函数) W
 - energy difference from Fermi level to vacuum
 - for metals, $W = q\chi$



p-type and n-type semiconductor

p-Si

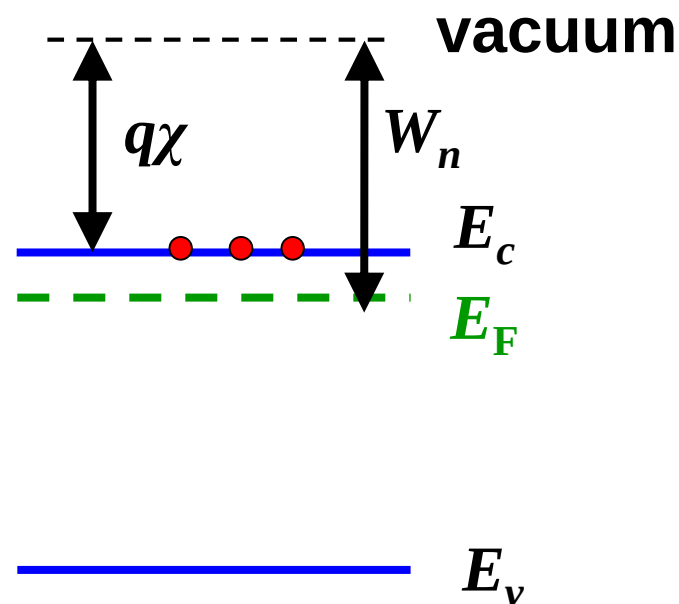


$$p_v = N_A$$

$$n_c = n_i^2 / p_v$$

$$p_v = P_v(T) e^{-(\mu - E_v)/k_B T}$$

n-Si



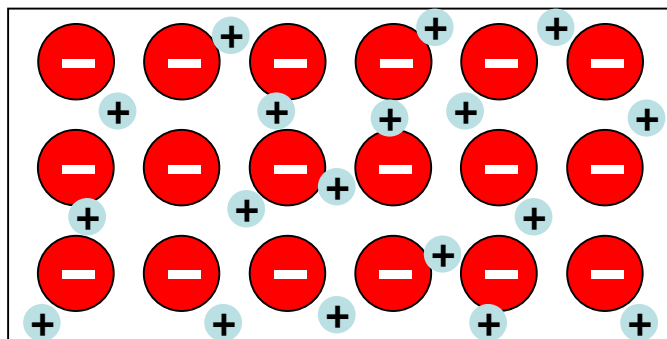
$$n_c = N_D$$

$$p_v = n_i^2 / n_c$$

$$n_c = N_c(T) e^{-(E_c - \mu)/k_B T}$$

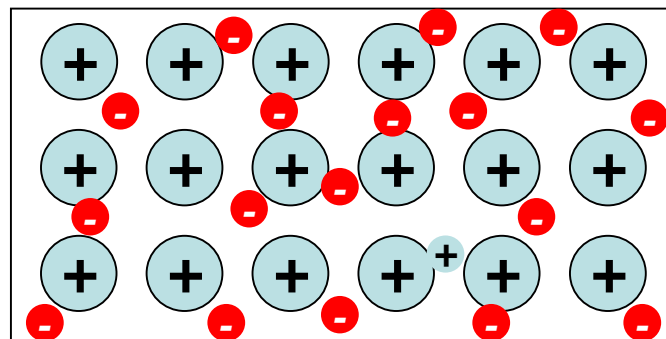
p-type and n-type semiconductor

p-Si



 acceptor  hole

n-Si



 donor  electron

$$p_v = N_A$$

$$n_c = n_i^2 / p_v$$

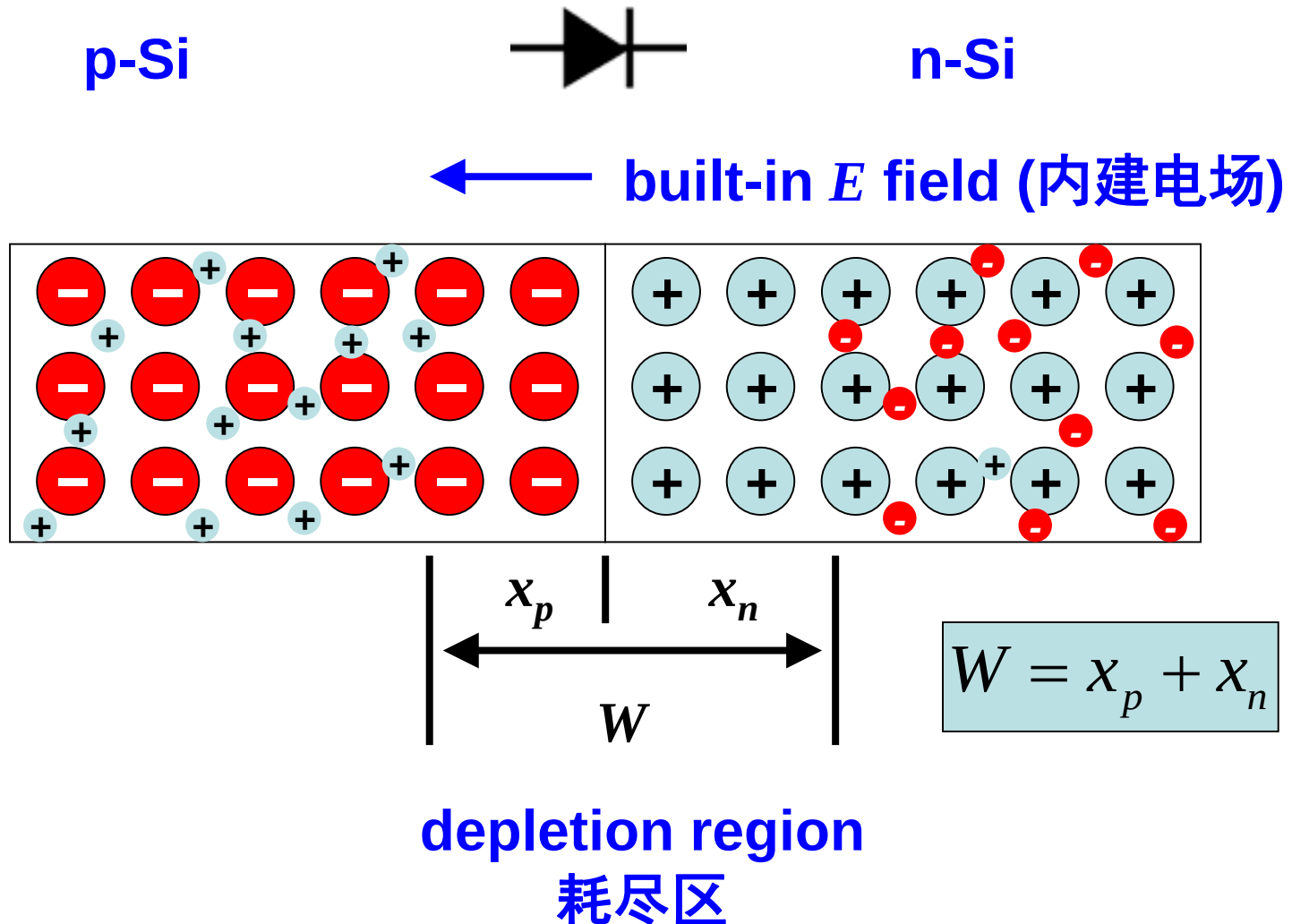
$$p_v = P_v(T) e^{-(\mu - E_v)/k_B T}$$

$$n_c = N_D$$

$$p_v = n_i^2 / n_c$$

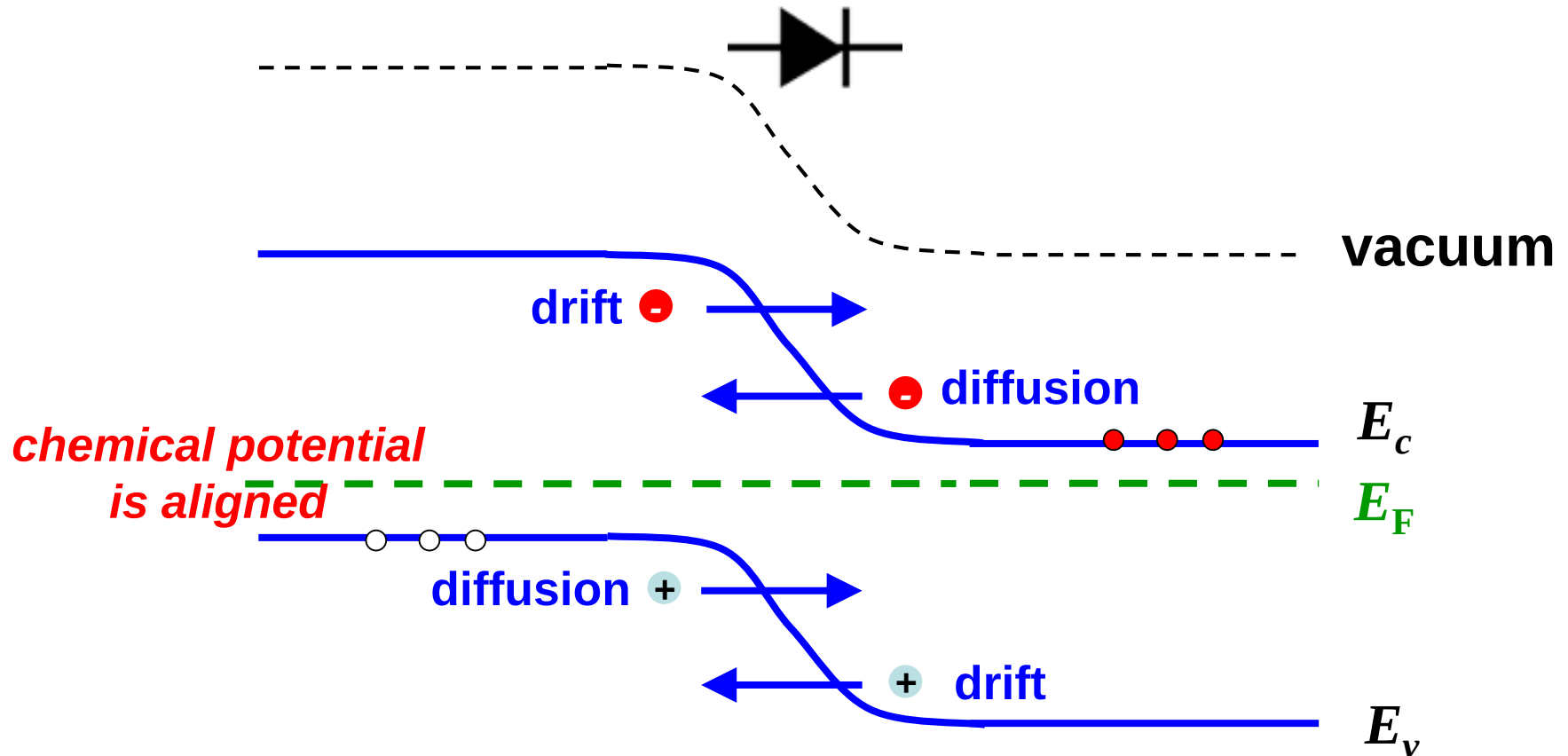
$$n_c = N_c(T) e^{-(E_c - \mu)/k_B T}$$

pn homojunction 同质结



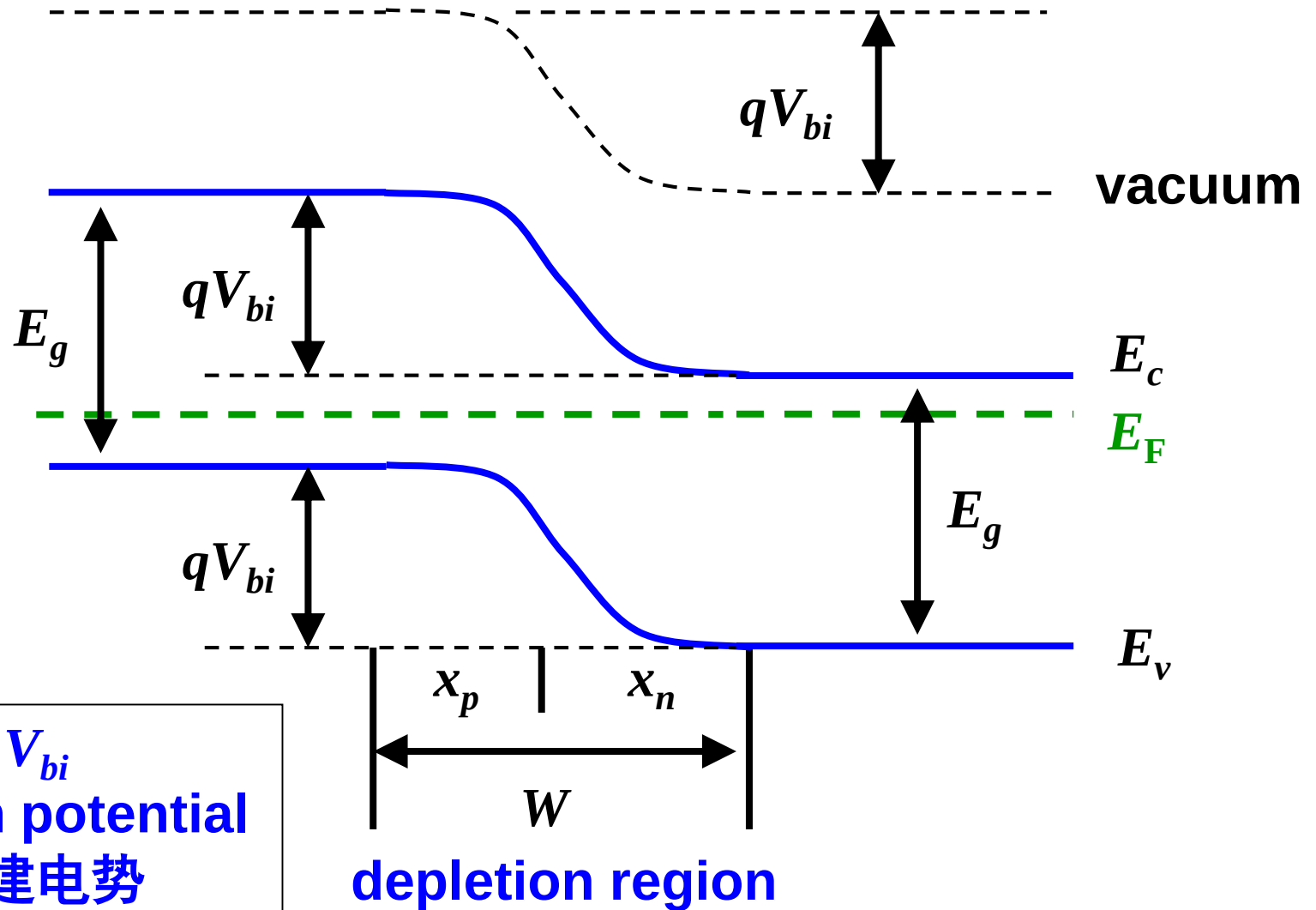
electrons and holes recombine

pn homojunction 同质结

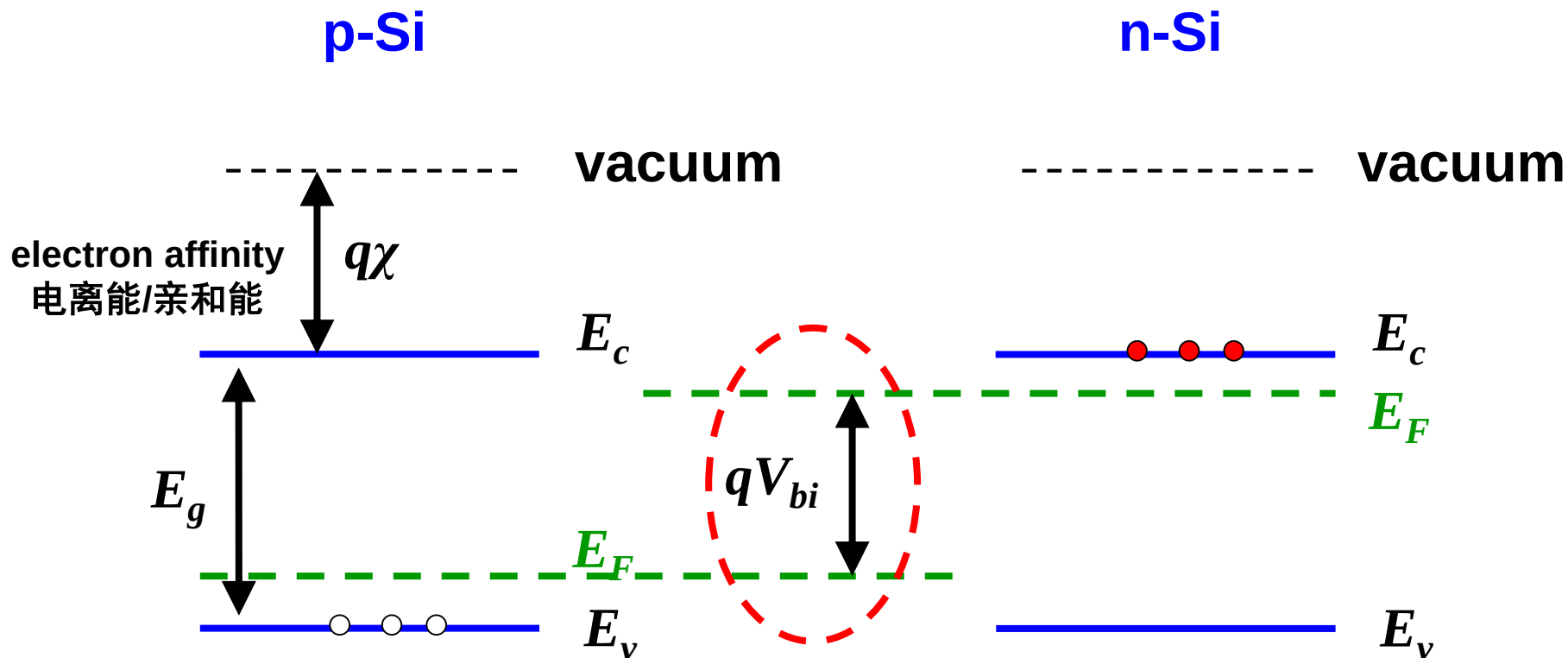


at thermal equilibrium, carrier diffusion is balanced by drift caused by the built-in field. Overall current = 0

pn homojunction 同质结



p-type and n-type semiconductor



$$p_v = N_A$$

$$n_c = n_i^2 / p_v$$

$$p_v = P_v(T) e^{-(\mu - E_v)/k_B T}$$

$$n_c = N_D$$

$$p_v = n_i^2 / n_c$$

$$n_c = N_c(T) e^{-(E_c - \mu)/k_B T}$$

qV_{bi} - built-in potential 内建电势

$$qV_{bi} = \mu_n - \mu_p = k_B T \cdot \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

Homework 7.6

q = electron charge 1.6×10^{-19} C

Example:

For a Si pn junction, if $N_A = 1 \text{e}18 \text{ cm}^{-3}$, $N_D = 1 \text{e}15 \text{ cm}^{-3}$,
and $n_i = 1.5 \text{e}10 \text{ cm}^{-3}$, $T = 300 \text{ K}$



$$V_{bi} = 0.75 \text{ V}$$

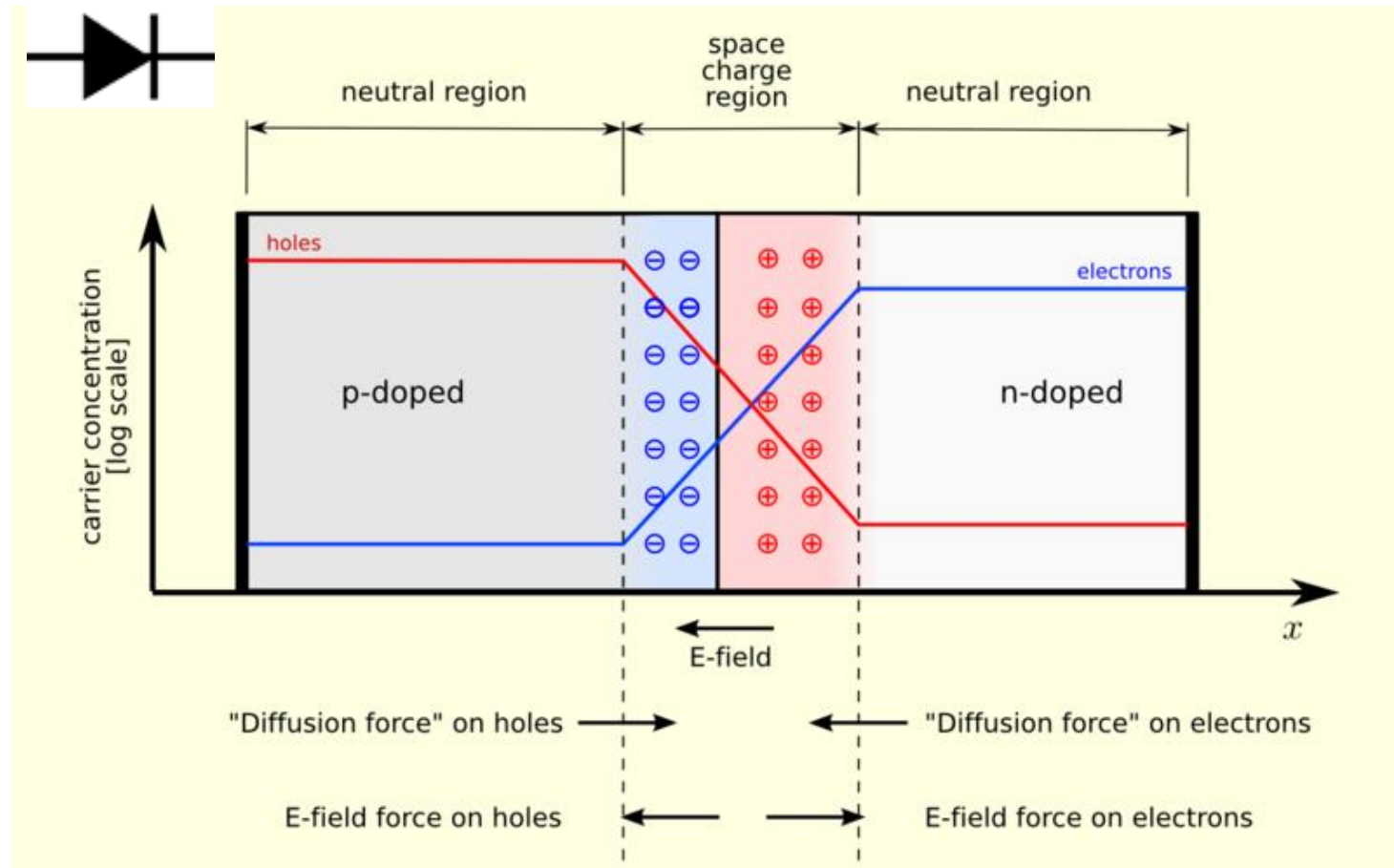
built-in voltage

$$qV_{bi} = 0.75 \text{ eV}$$

<

$$E_g = 1.12 \text{ eV}$$

pn homojunction 同质结



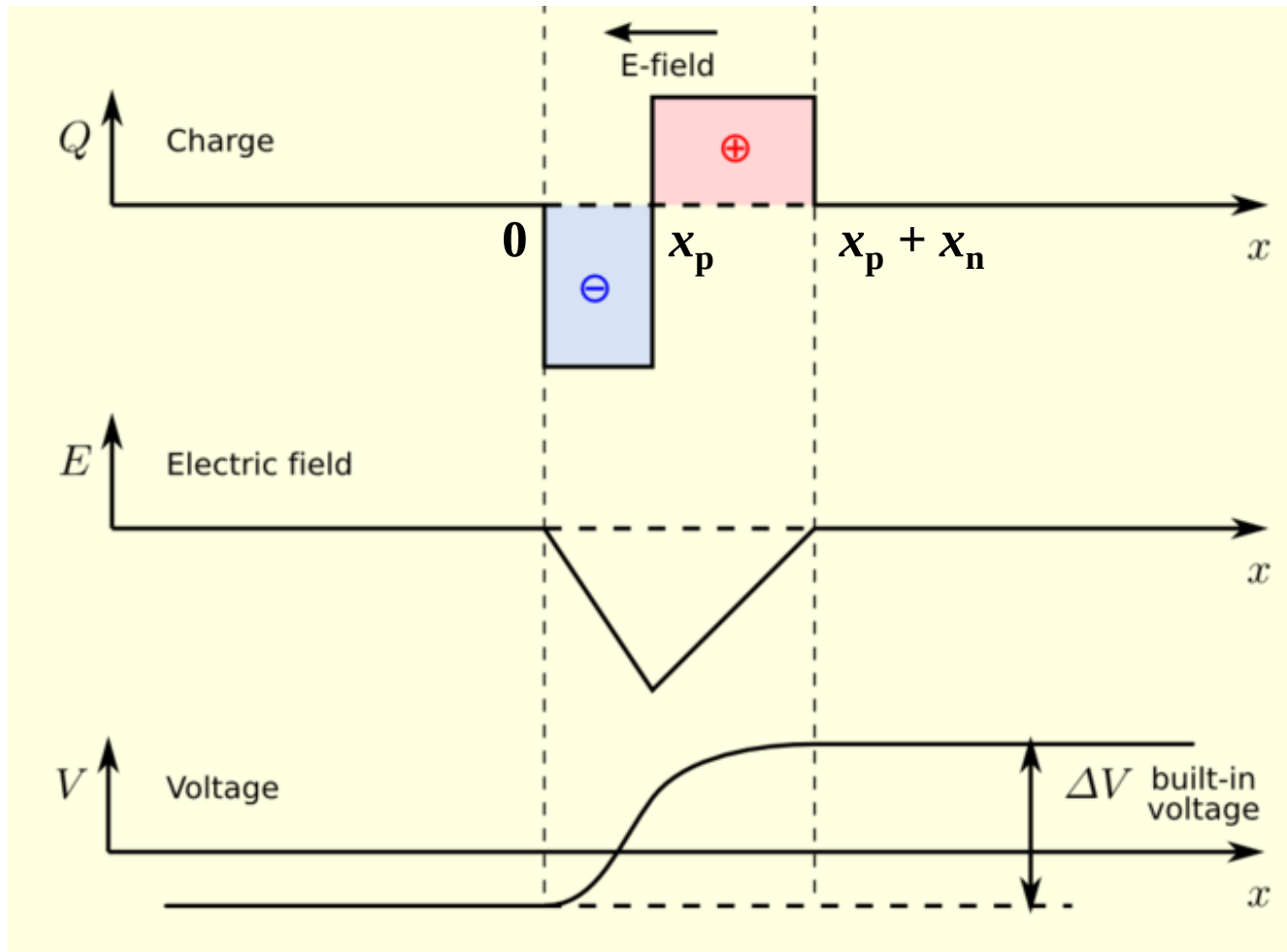
$$W = x_p + x_n$$

$$W = x_p + x_n$$

Q: How to calculate the depletion width W ?

Full-depletion Approximation

Assume abrupt transition



Charge Balance

$$N_A x_p = N_D x_n$$

Gauss's Law

$$\frac{\partial E}{\partial x} = \frac{q}{\epsilon_s} Q(x)$$

$$\frac{\partial V}{\partial x} = -E(x)$$

Full-depletion Approximation

Assume abrupt transition

and

$$V_{bi} = \frac{k_B T}{q} \cdot \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

Lecture Note 3.12



$$W = \sqrt{\frac{2\epsilon_s}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) V_{bi}}$$

$$\epsilon_s = \epsilon_0 \epsilon_r$$

$$x_p = \frac{N_D}{N_A + N_D} W$$

$$x_n = \frac{N_A}{N_A + N_D} W$$

ϵ_s - dielectric constant / permittivity (F/m) 介电常数

ϵ_r - relative dielectric constant

q - electron charge 1.6×10^{-19} C

Full-depletion Approximation

Example:

For a Si pn junction, if $N_A = 1\text{e}18\text{ cm}^{-3}$, $N_D = 1\text{e}15\text{ cm}^{-3}$,
and $n_i = 1.5\text{e}10\text{ cm}^{-3}$, $T = 300\text{ K}$, $\epsilon_r = 11.2$

→ $V_{bi} = 0.75\text{ V}$

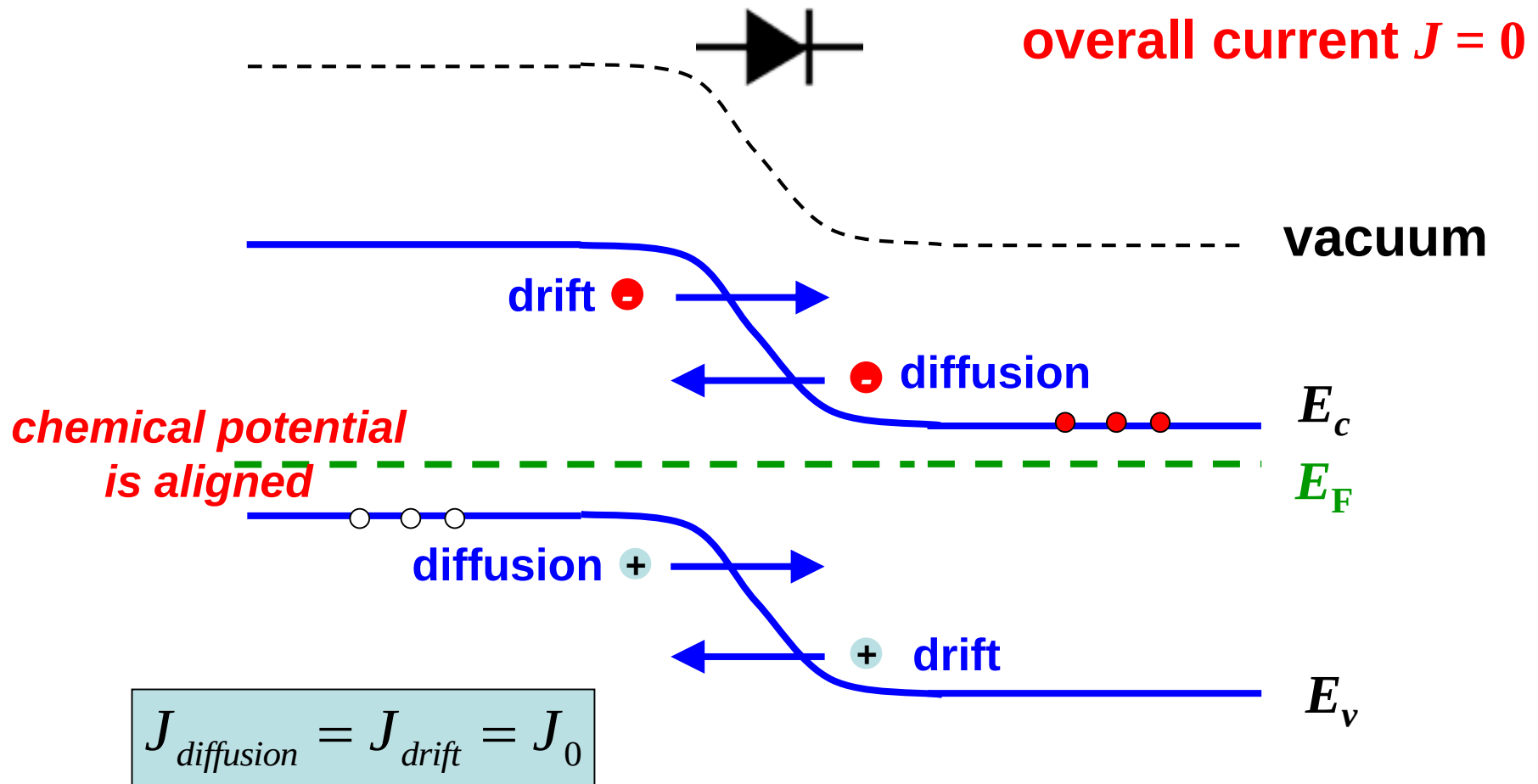
→ $W = \sqrt{\frac{2\epsilon_s}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) V_{bi}} = 964\text{ nm}$

$$x_p = 0.9\text{ nm}$$

$$x_n = 963\text{ nm}$$

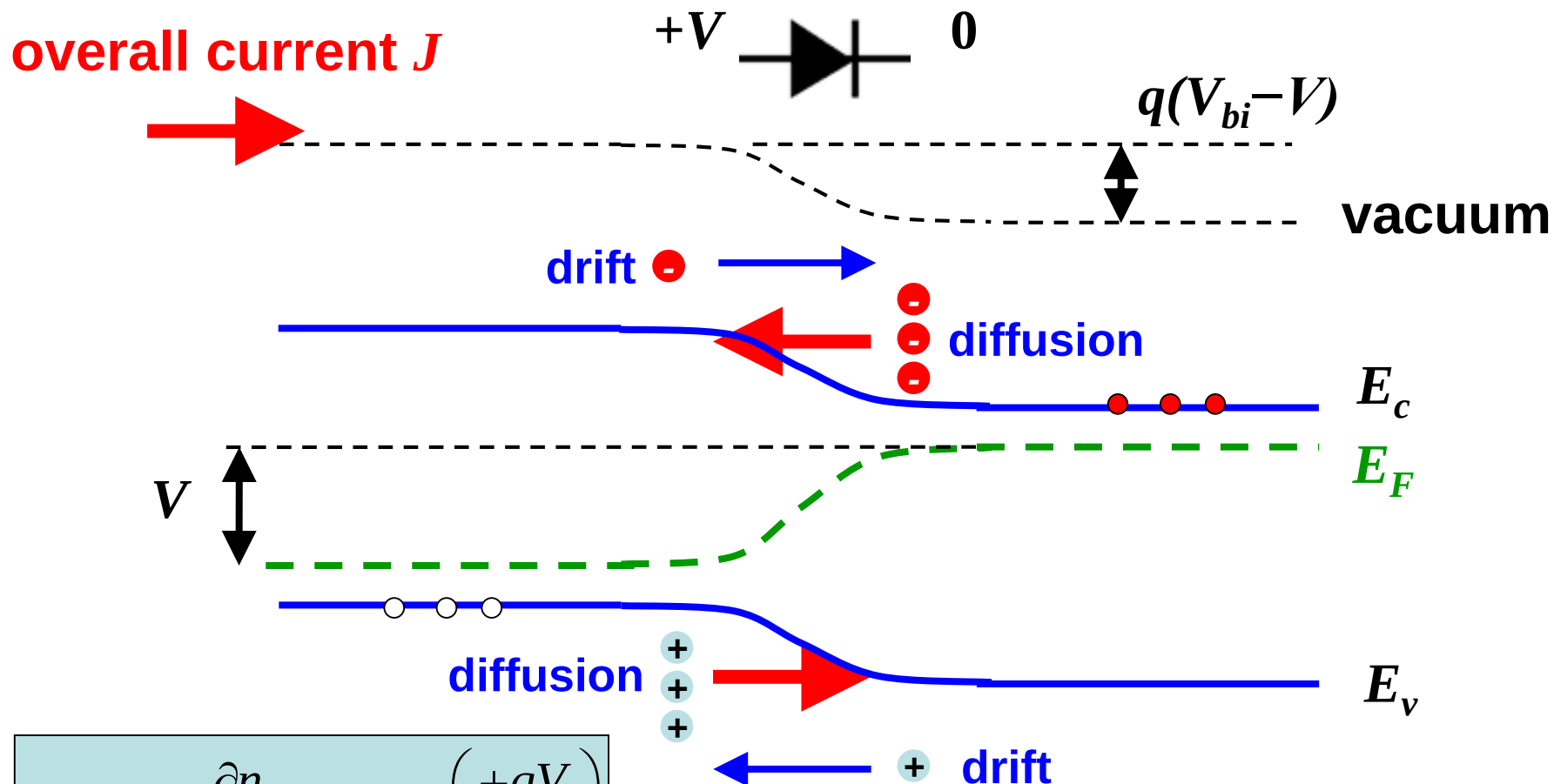
*higher doping
-> smaller width*

At Equilibrium ($V = 0$)



at thermal equilibrium, carrier diffusion is balanced by drift caused by the built-in field. Overall current = 0

At Forward Bias ($p \rightarrow n$)



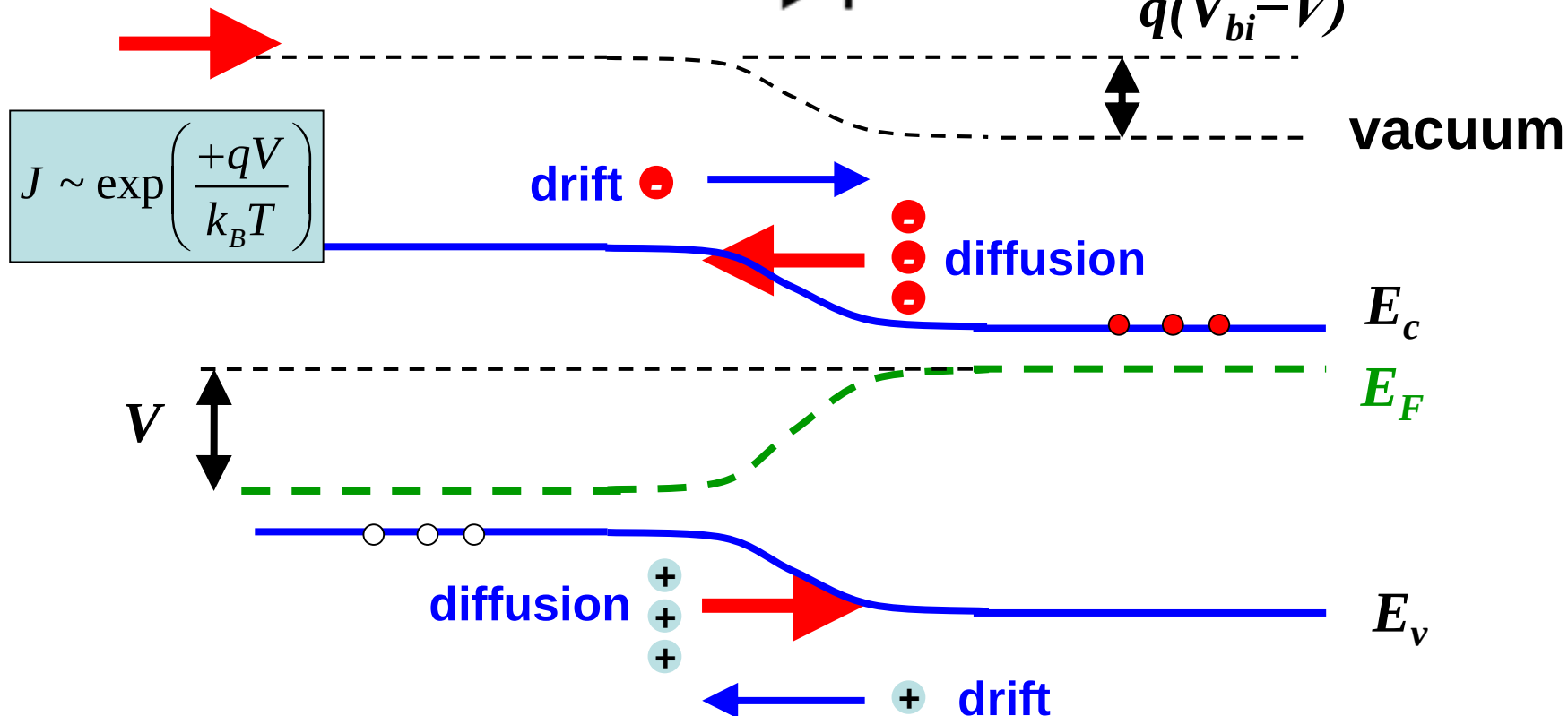
$$J_{diffusion} \sim \frac{\partial n}{\partial x} \sim J_0 \exp\left(\frac{+qV}{k_B T}\right)$$

$$J_{drift} \approx J_0$$

$$J = J_{diffusion} - J_{drift} = J_0 \left[\exp\left(\frac{+qV}{k_B T}\right) - 1 \right]$$

At Forward Bias ($p \rightarrow n$)

overall current J

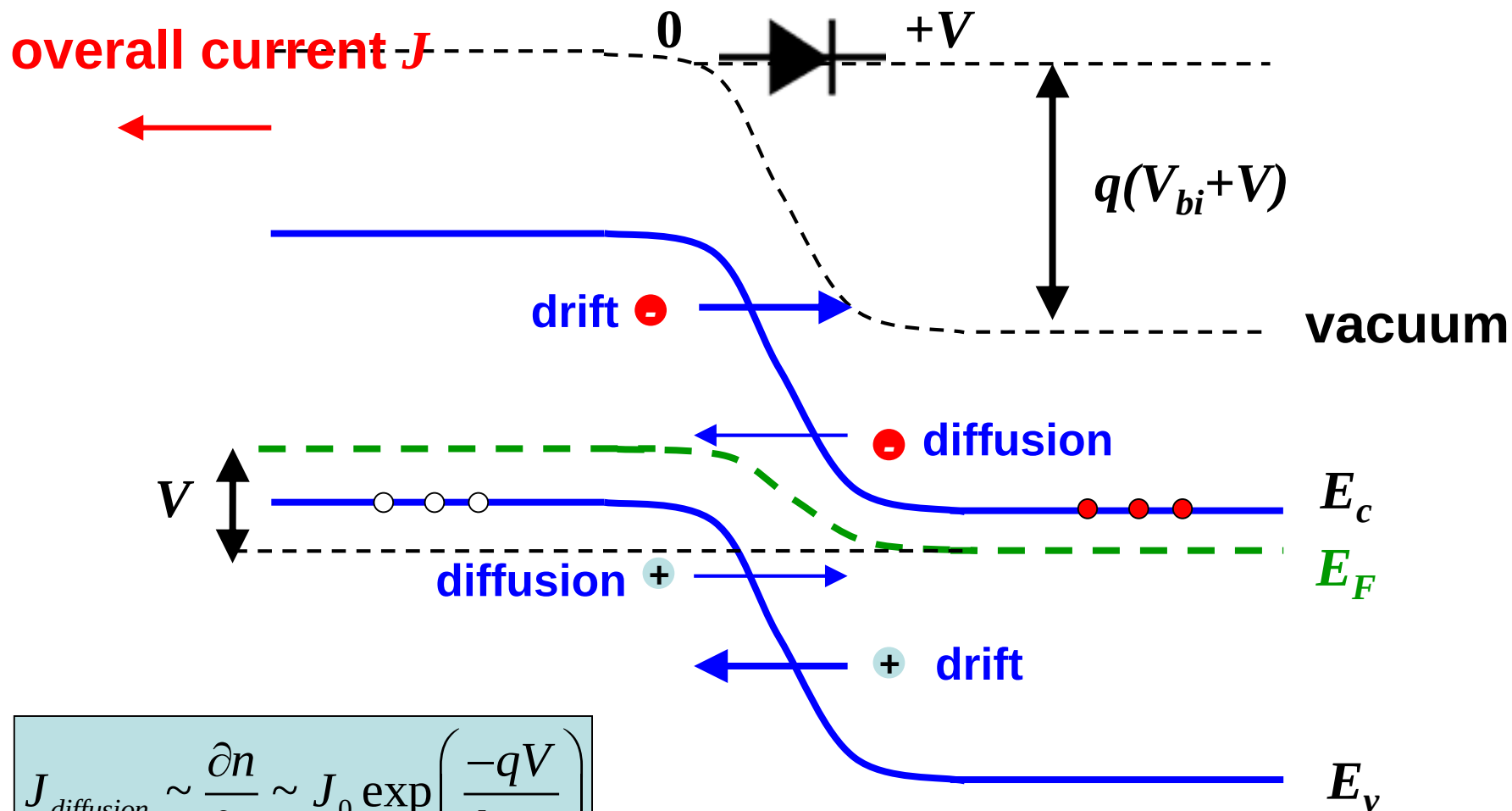


$$J \sim \exp\left(\frac{+qV}{k_B T}\right)$$

$$W = \sqrt{\frac{2\epsilon_s}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) (V_{bi} - V)}$$

Depletion region decreases

At Reverse Bias ($p \leftarrow n$)

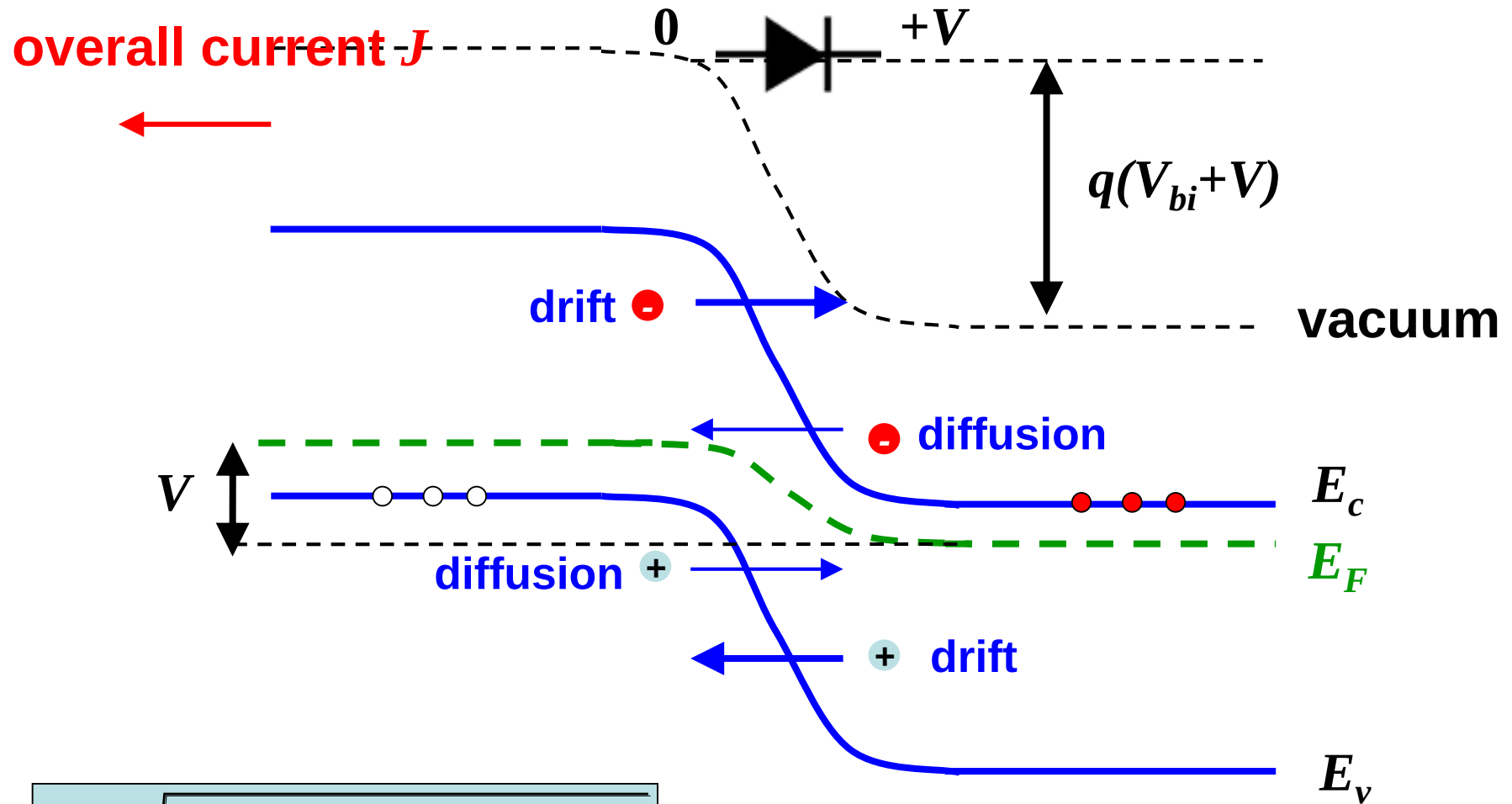


$$J_{diffusion} \sim \frac{\partial n}{\partial x} \sim J_0 \exp\left(\frac{-qV}{k_B T}\right)$$

$$J_{drift} \approx J_0$$

$$J = J_{diffusion} - J_{drift} = J_0 \left[\exp\left(\frac{-qV}{k_B T}\right) - 1 \right]$$

At Reverse Bias ($p \leftarrow n$)

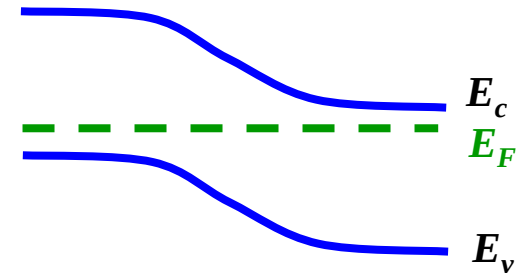
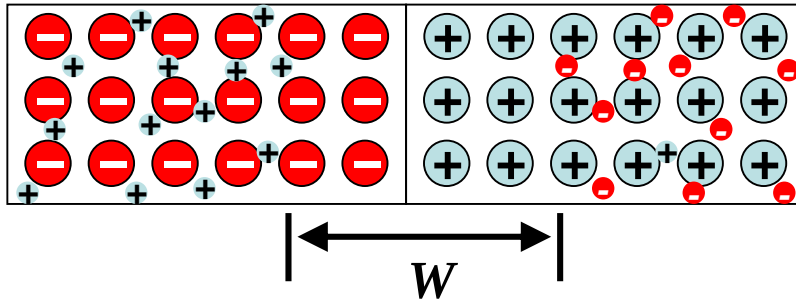


$$W = \sqrt{\frac{2\epsilon_s}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) (V_{bi} + V)}$$

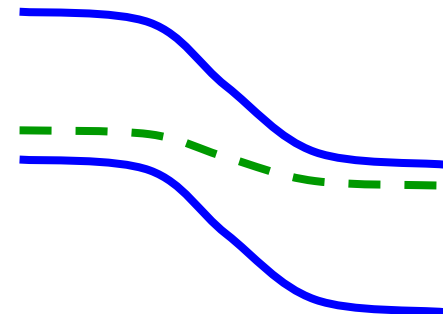
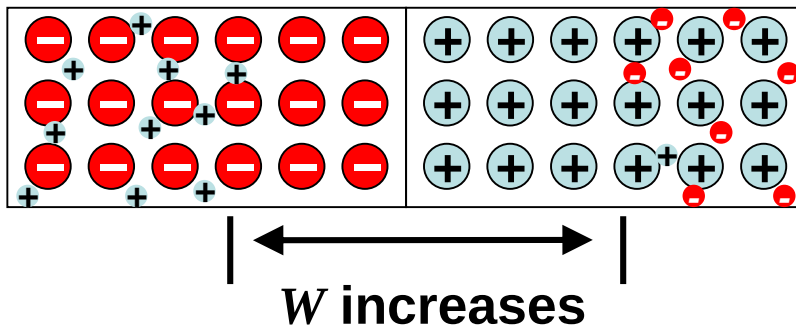
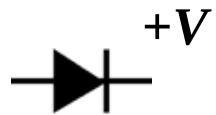
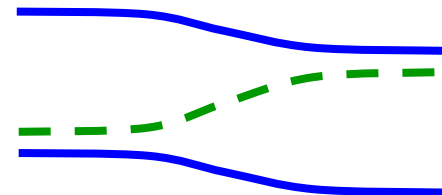
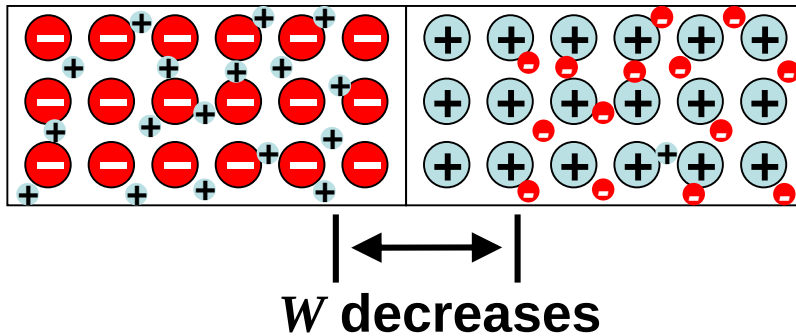
Depletion region increases

Summary

$V = 0$



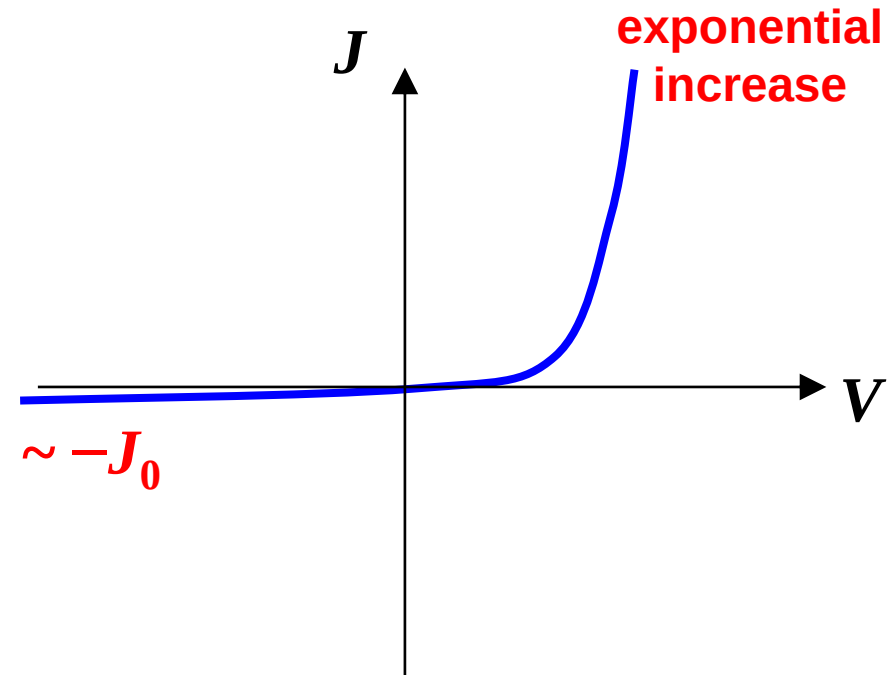
$+V$



Current-Voltage Relation

pn junction - diode 二极管

$$J = J_0 \left[\exp\left(\frac{qV}{nk_B T}\right) - 1 \right]$$



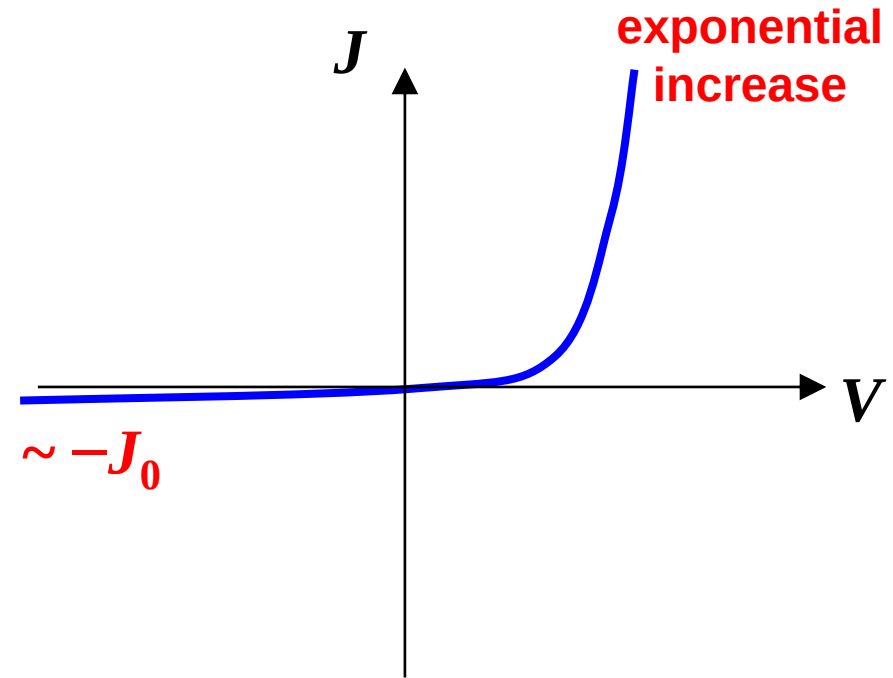
J_0 - dark/leakage/saturation current density
depend on bandgap, defects, temperature, ...

n - ideality factor (for ideal case, $n = 1$)

Current-Voltage Relation

pn junction - diode 二极管

$$J = J_0 \left[\exp\left(\frac{qV}{nk_B T}\right) - 1 \right]$$



ideal diode model

$$J_0 \approx q \frac{D_n}{L_n} \frac{n_i^2}{N_A} + q \frac{D_p}{L_p} \frac{n_i^2}{N_D}$$

D - diffusivity (m^2/s) 扩散系数

τ - carrier lifetime (s)

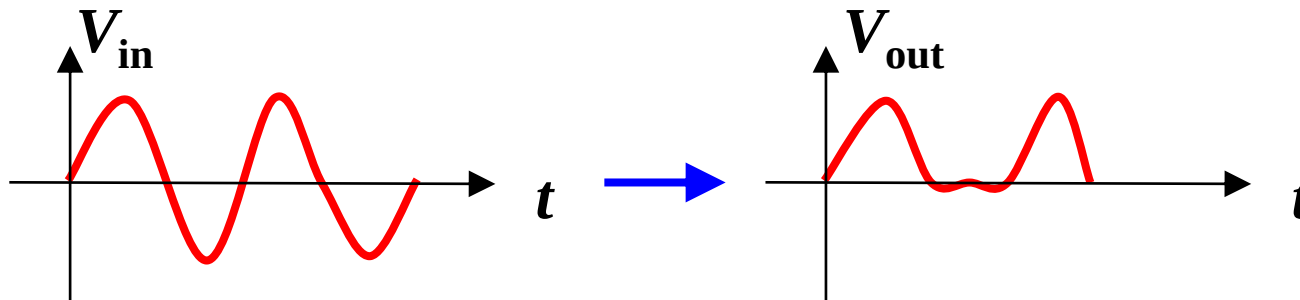
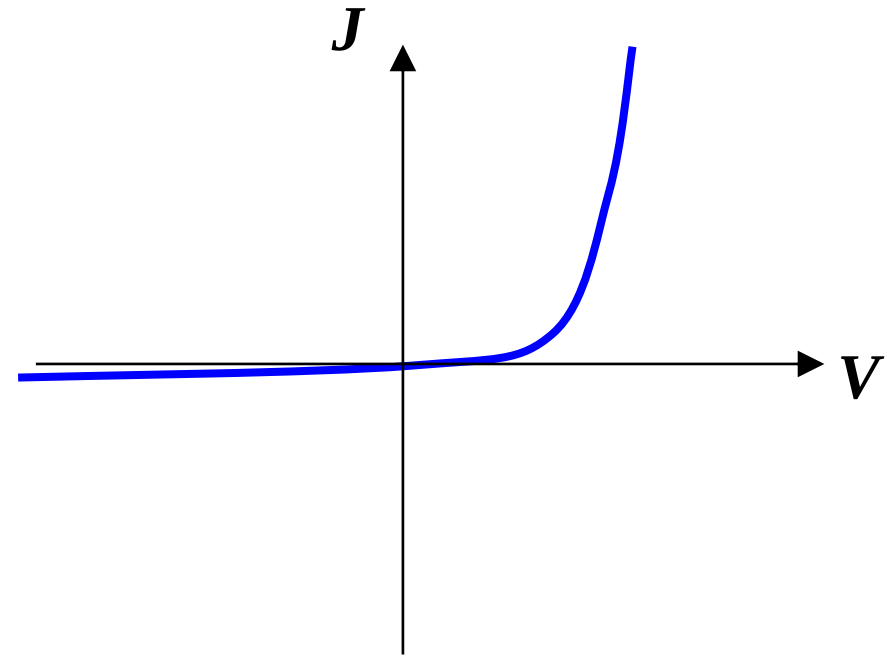
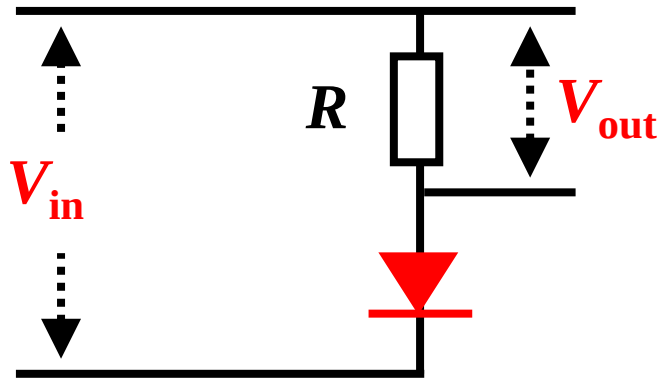
L - diffusion length (m)

$$L = \sqrt{D\tau}$$

Current-Voltage Relation

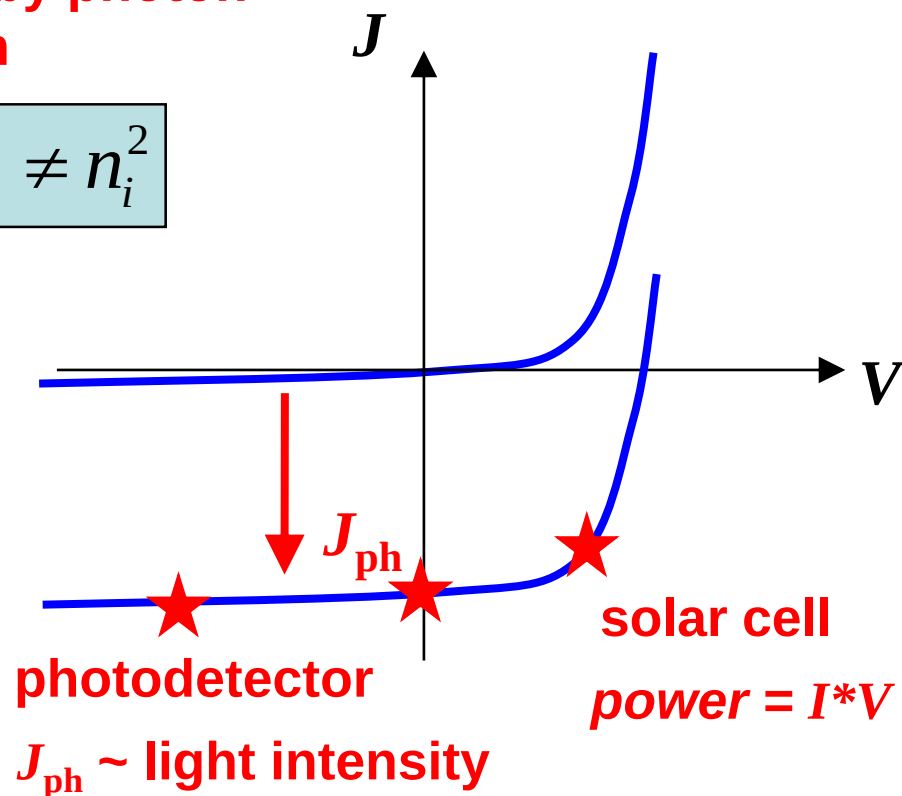
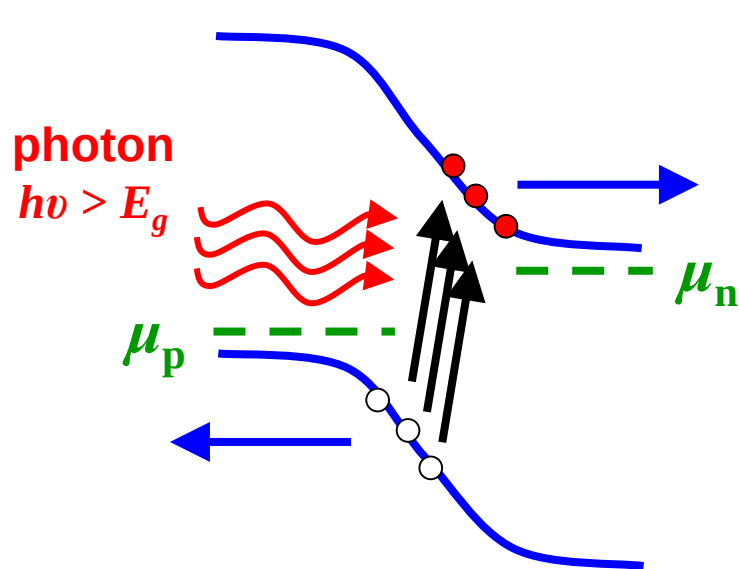
pn junction - diode 二极管

rectifier 整流管



Solar Cell / Photodetector

extra carriers generated by photon
non-equilibrium

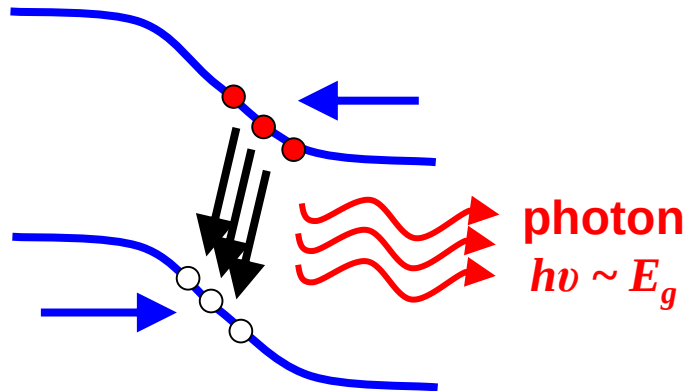


$$J = J_0 \left[\exp\left(\frac{qV}{nk_B T}\right) - 1 \right] - J_{ph}$$

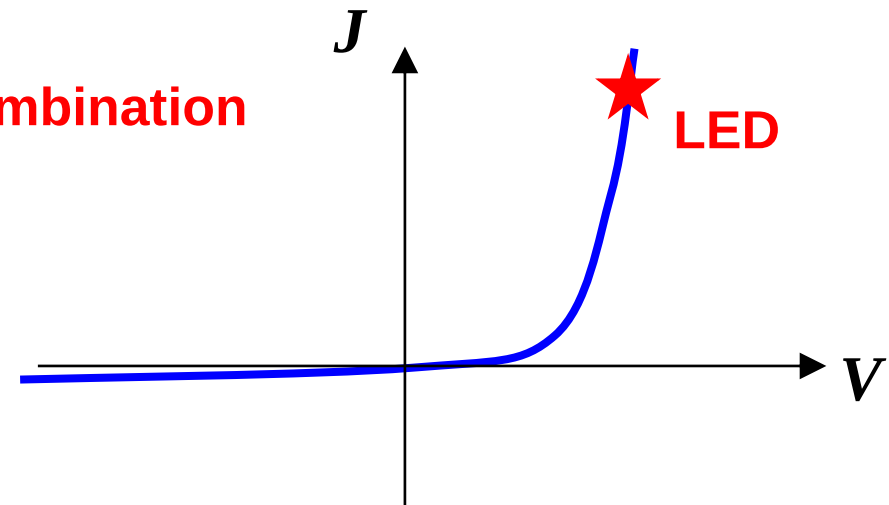
photocurrent

Light-Emitting Diode (LED)

at forward bias
photon generation by radiative recombination



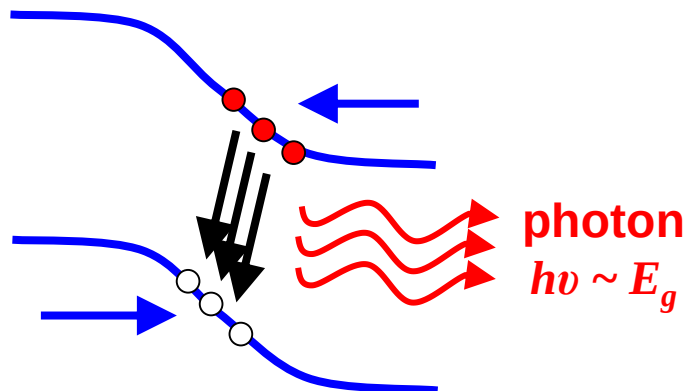
$$J = J_0 \left[\exp\left(\frac{qV}{nk_B T}\right) - 1 \right]$$



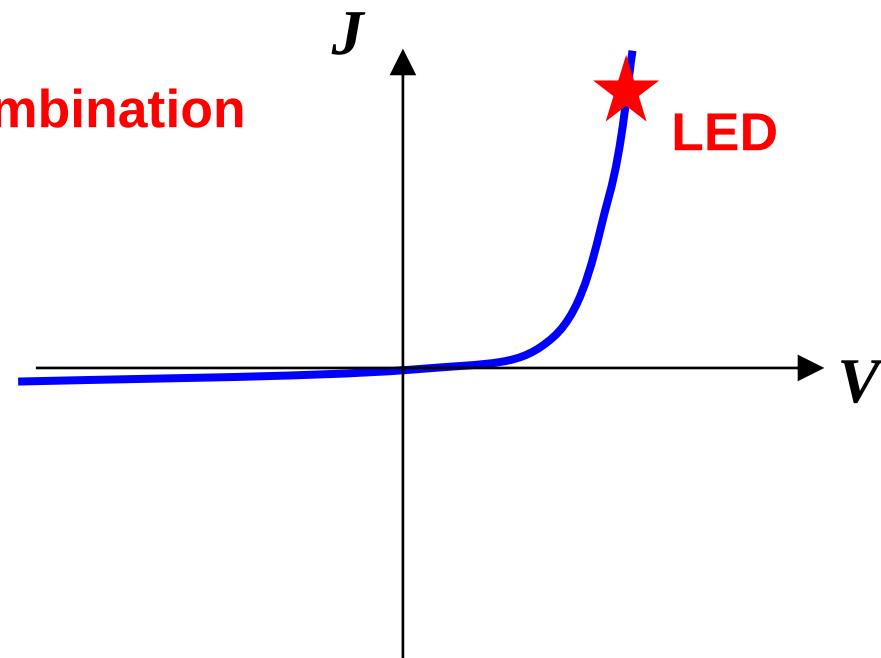
LEDs

Light-Emitting Diode (LED)

at forward bias
photon generation by radiative recombination



$$J = J_0 \left[\exp\left(\frac{qV}{nk_B T}\right) - 1 \right]$$

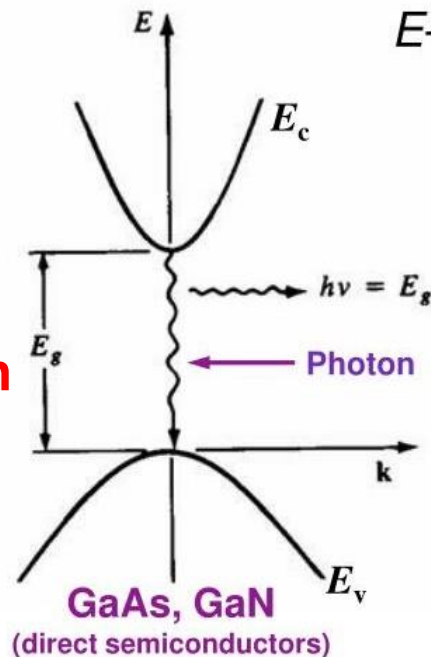


optical power $\sim J \cdot \eta$

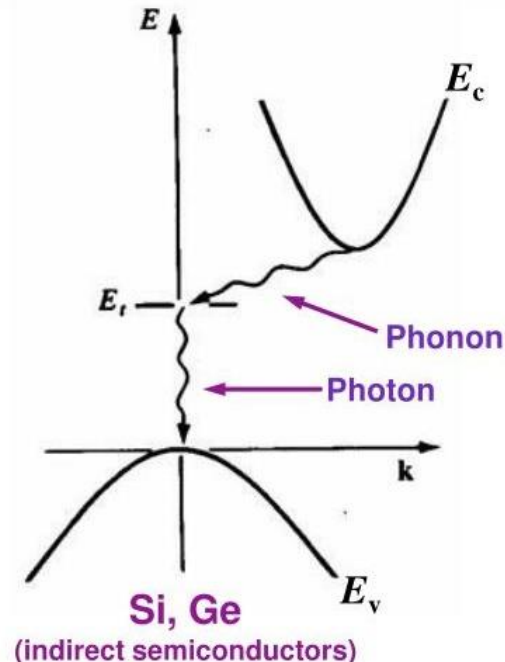
η - conversion efficiency
 $\eta < 100\%$, because of non-radiative recombination (generating heat)

Light Emission Efficiency

radiative
recombination



- Little change in momentum is required for recombination
- Momentum is conserved by photon (light) emission

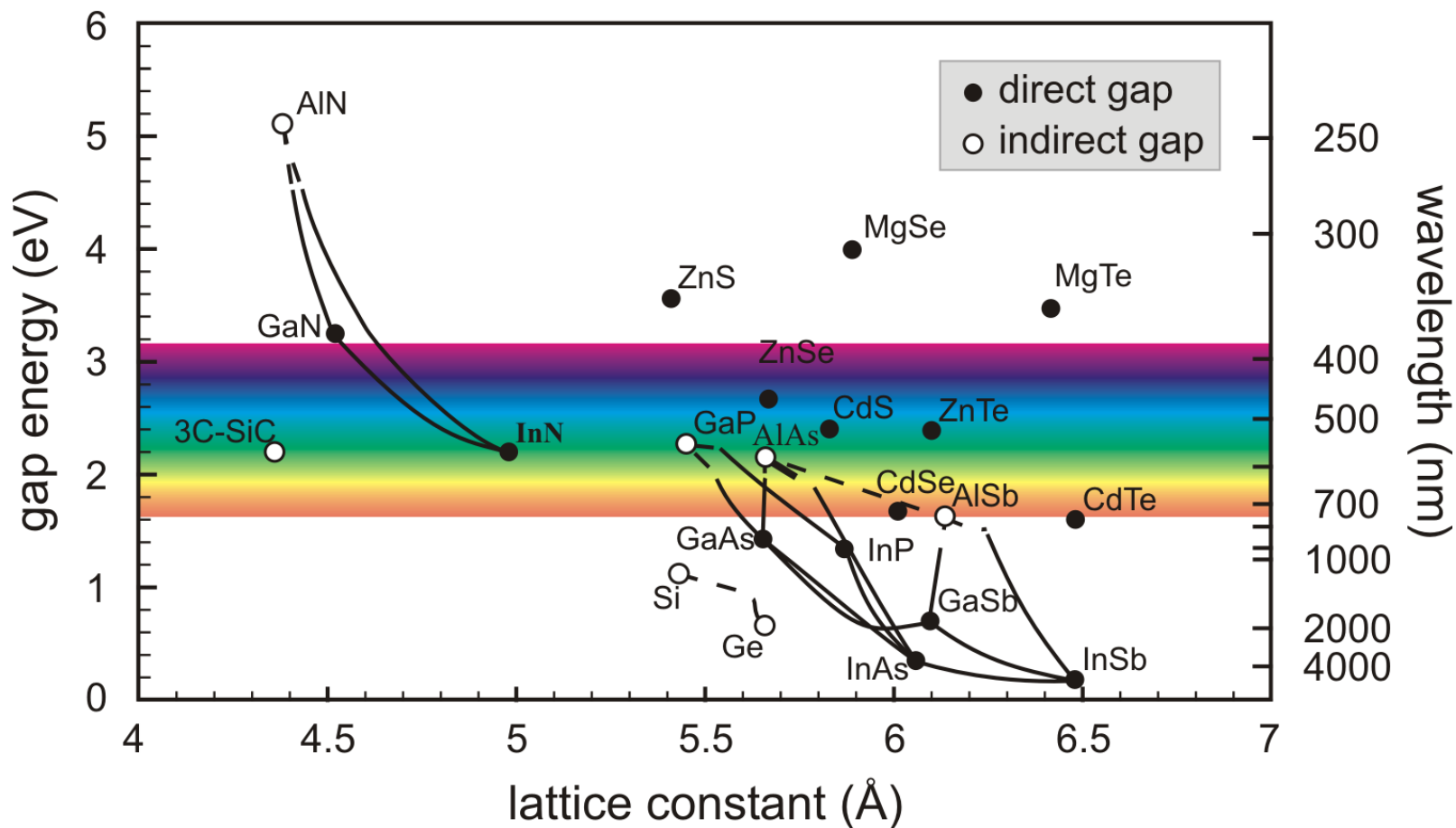


- Large change in momentum is required for recombination
- Momentum is conserved by mainly phonon (vibration) emission + photon emission

Direct bandgap semiconductors like GaAs, GaN are more suitable for LEDs and lasers, more radiative recombinations

Indirect bandgap semiconductors like Si, Ge do not emit light efficiently more non-radiative recombinations

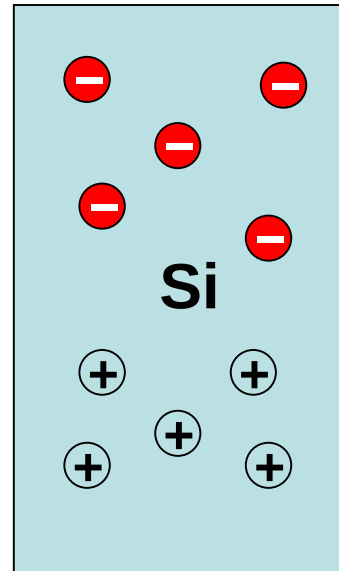
Materials Choices for Light Emission



Semiconductors with direct gaps are more suitable for LEDs and lasers

History of PN Junction

- Like most other parts in solid-state physics, the pn junction is first a scientific discovery, then an engineering design



In a melted Si crystal, light p-dopants (like B, Al) float in the top part, while heavy n-dopants (like P, As) sink in the bottom part

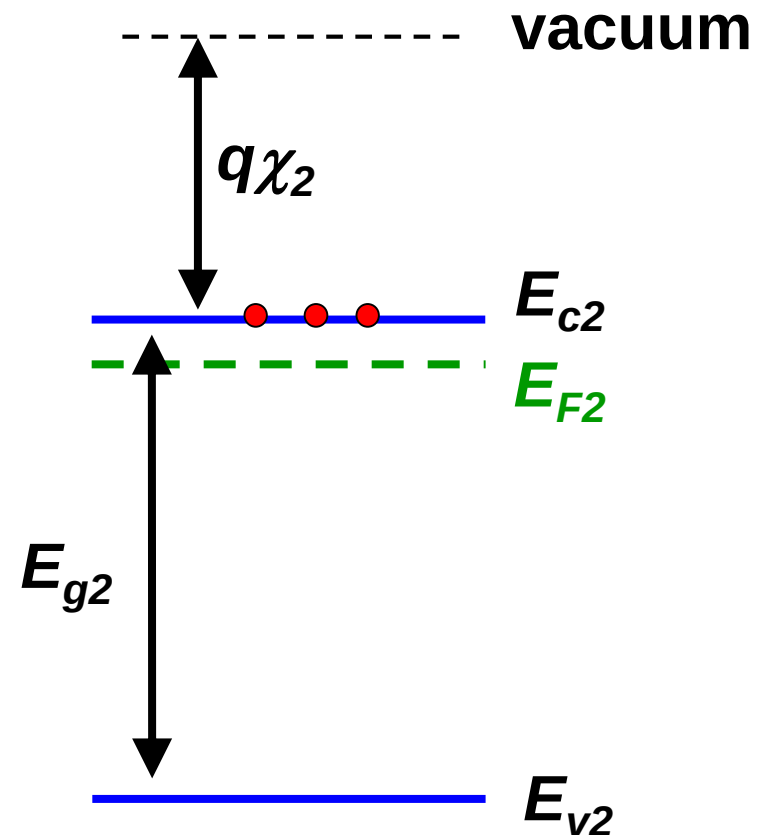
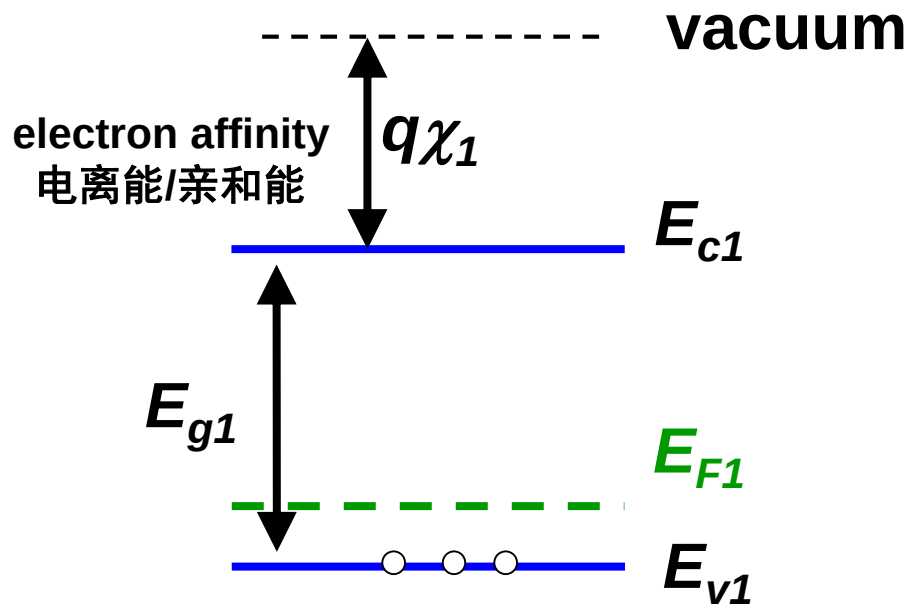
In 1940, Russell Ohl at Bell Labs accidentally created a Si pn junction.

Heterojunction 异质结

Case 1

p-Si

n-GaAs

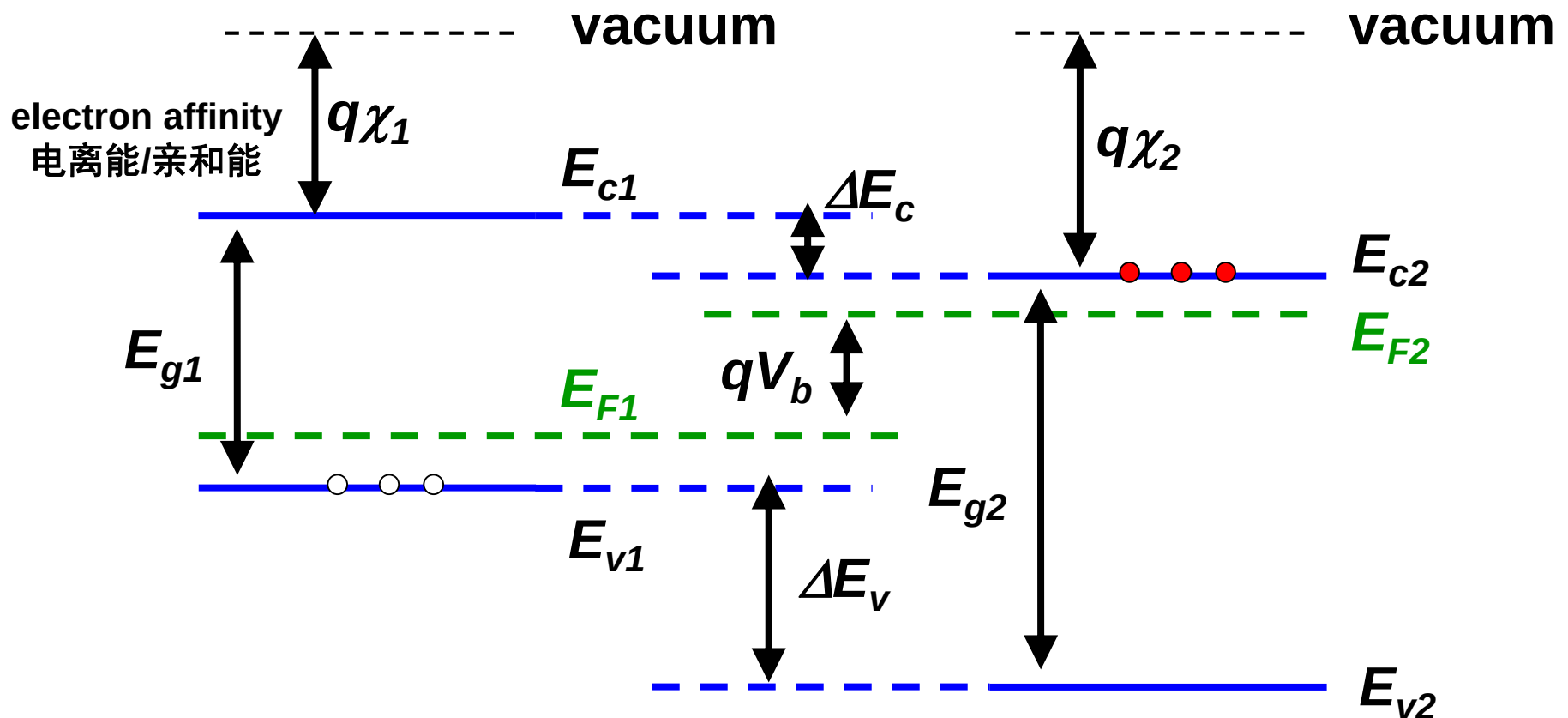


Heterojunction 异质结

Case 1

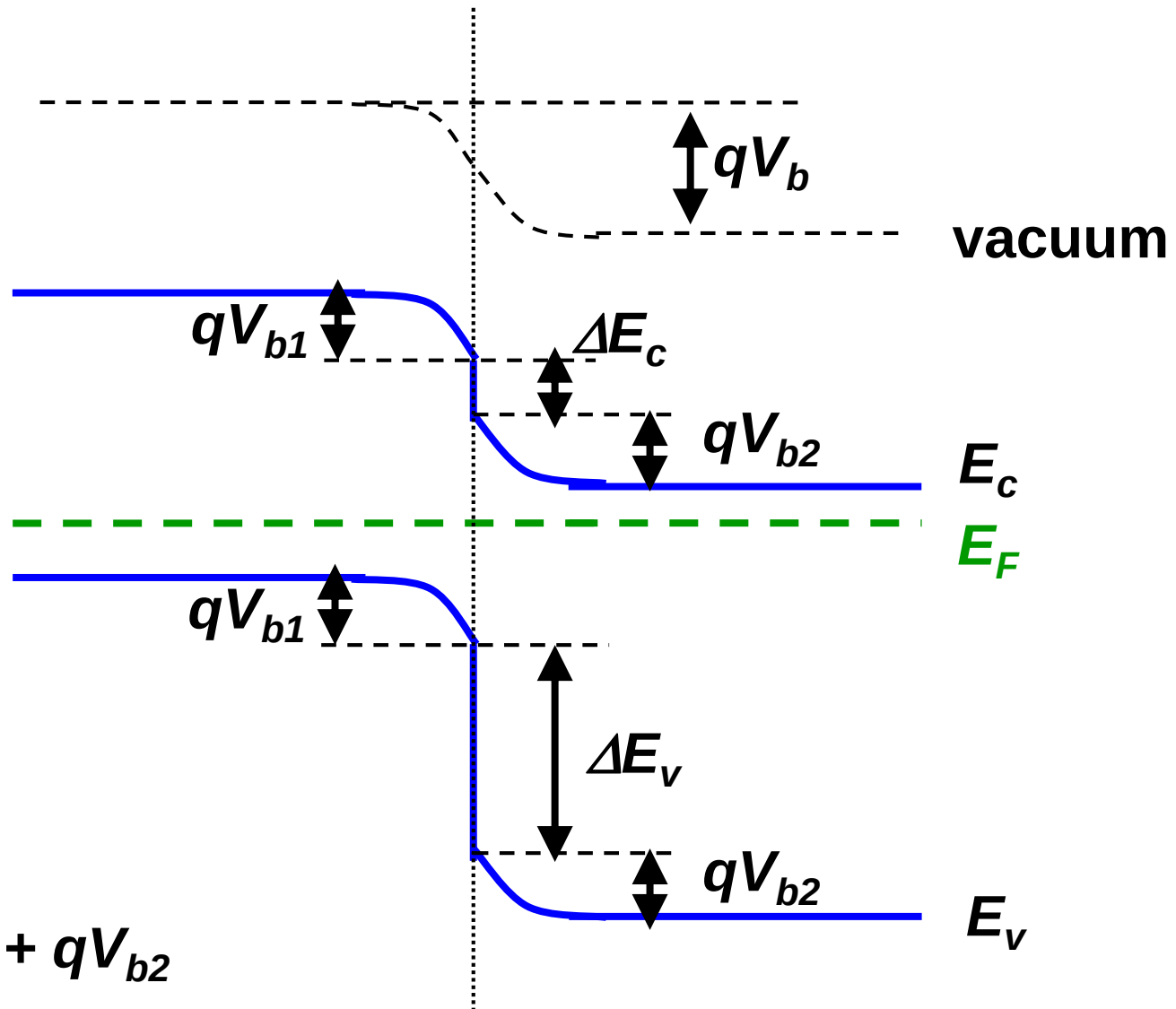
p-Si

n-GaAs



Heterojunction 异质结

Case 1

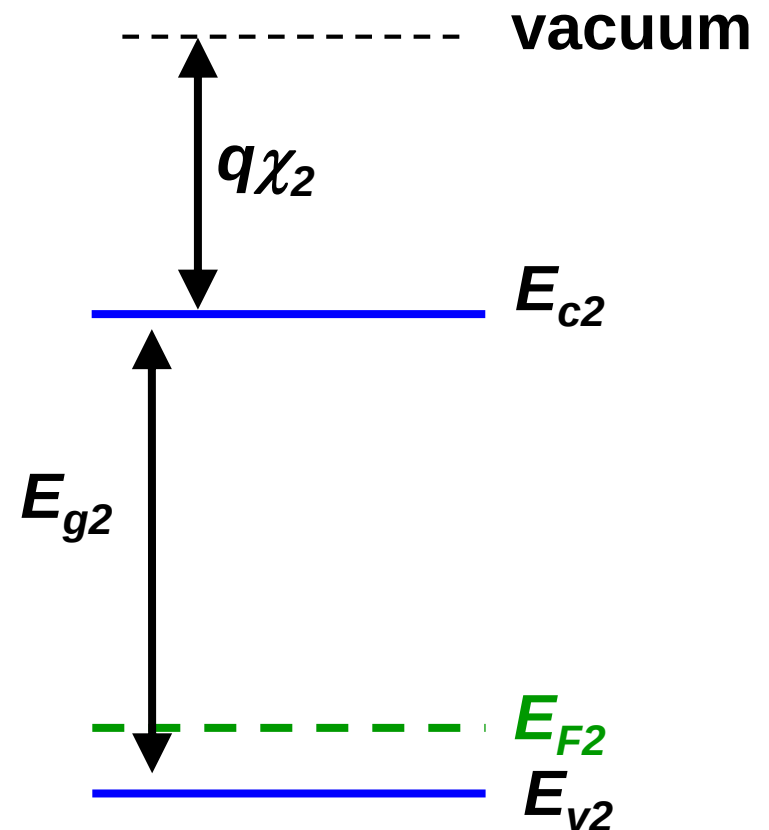
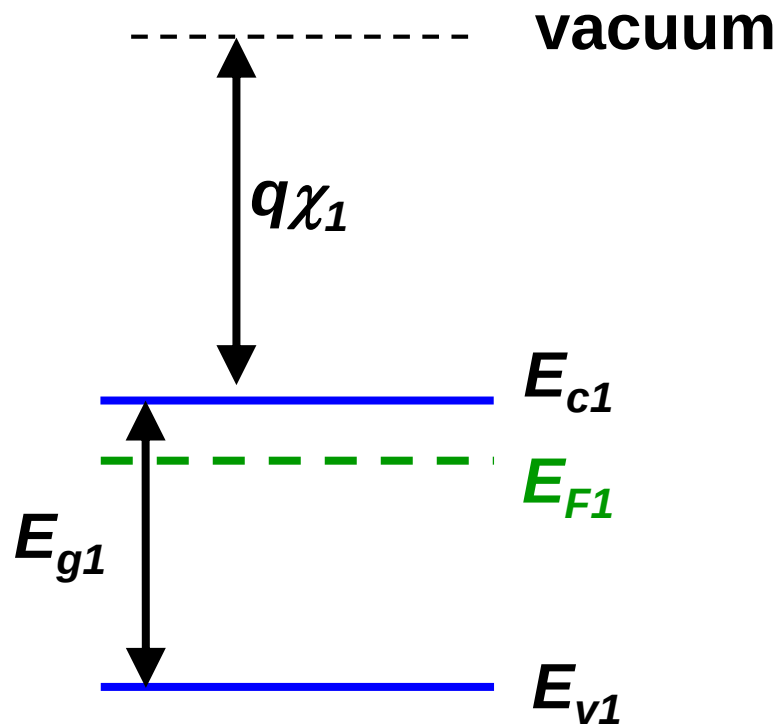


$$qV_b = qV_{b1} + qV_{b2}$$

Heterojunction 异质结

Case 2 n-InGaAs

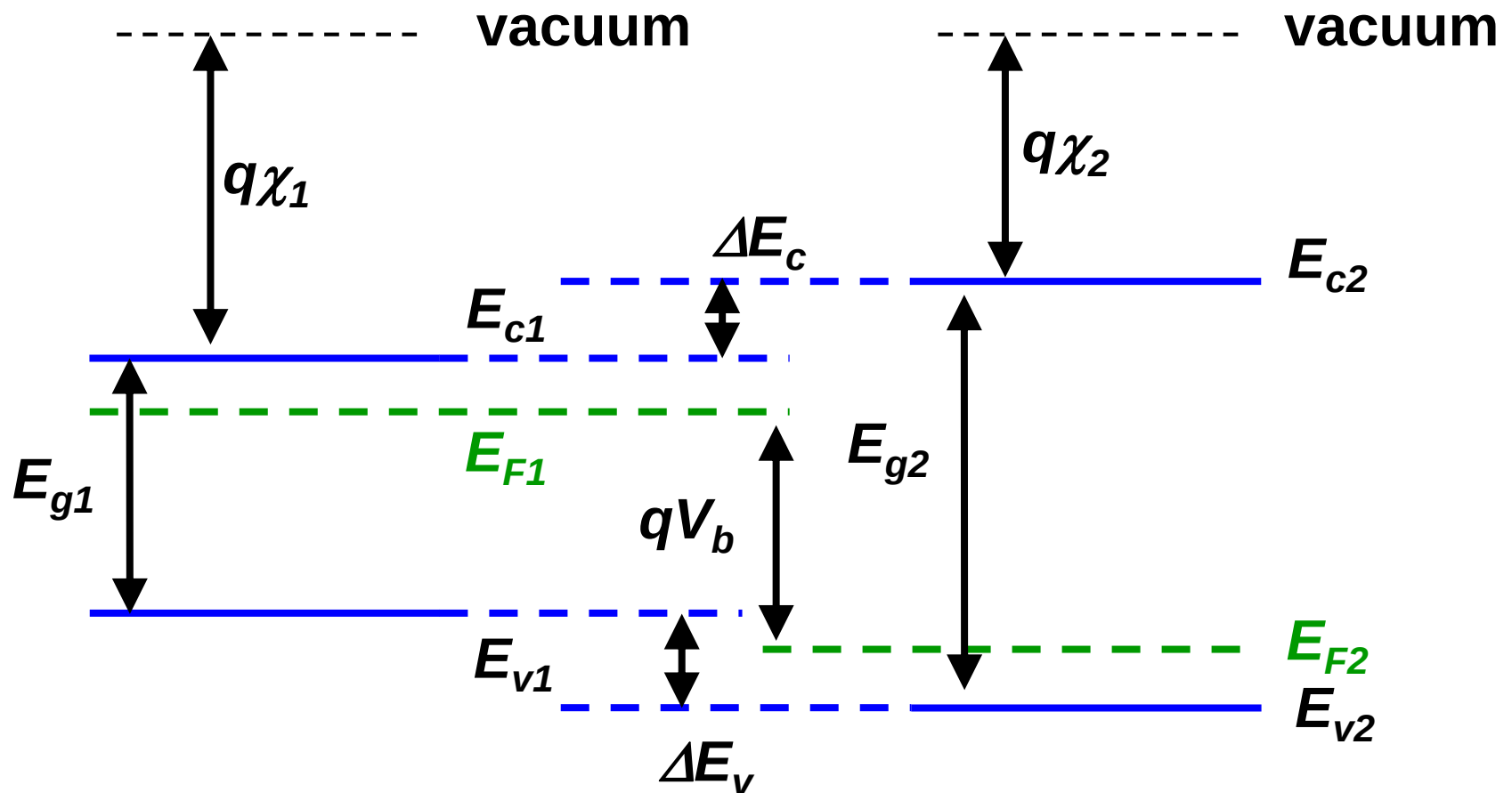
p-GaAs



Heterojunction 异质结

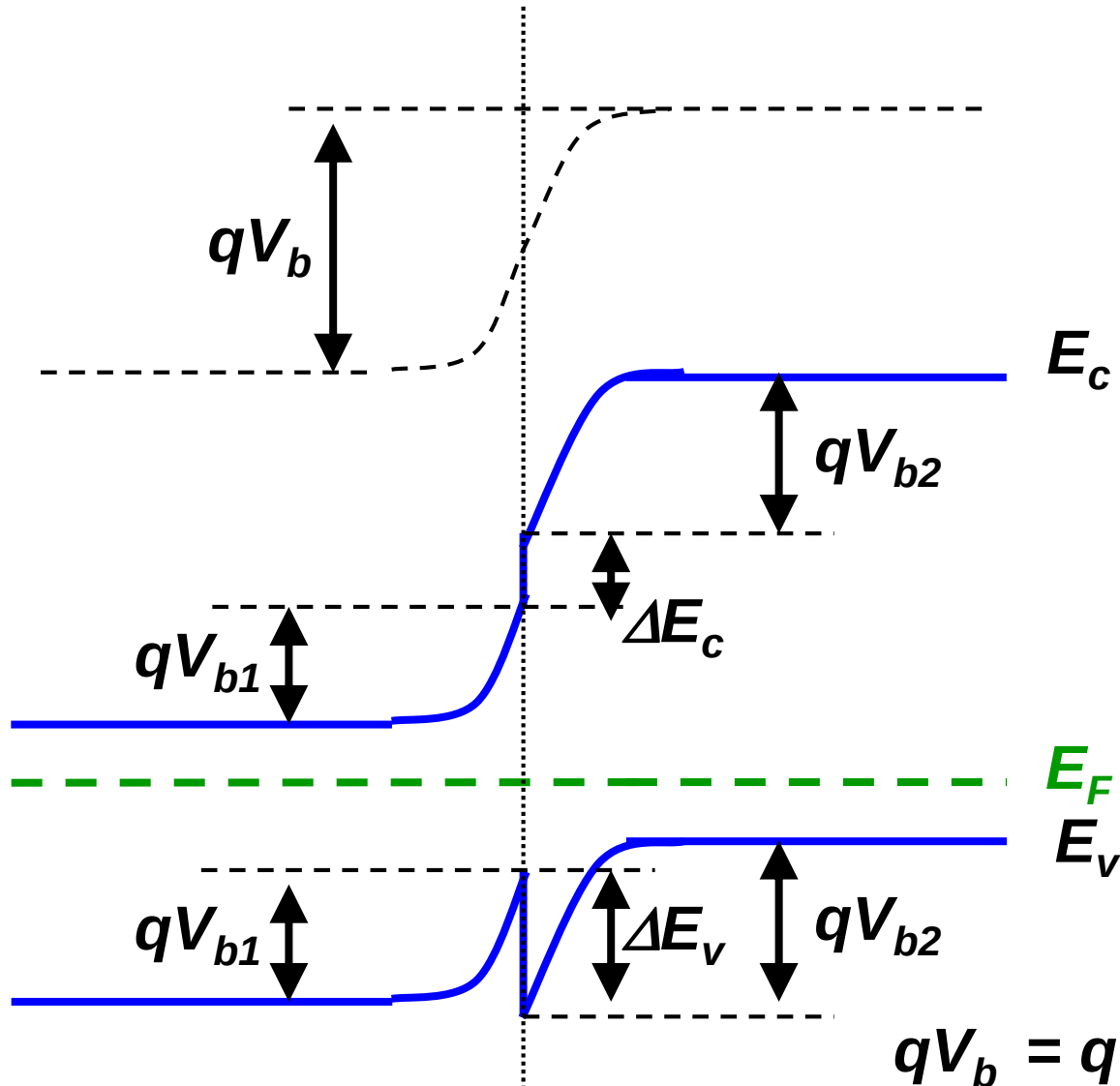
Case 2 n-InGaAs

p-GaAs

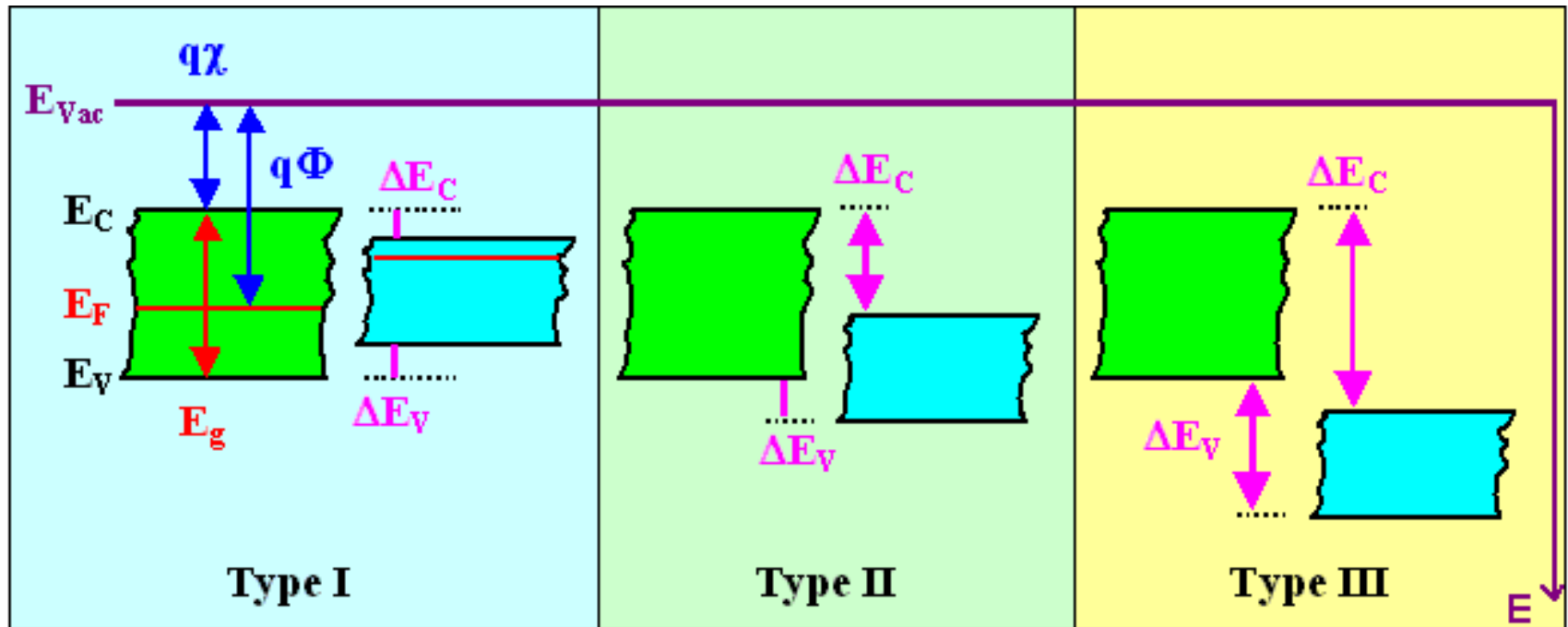


Heterojunction 异质结

Case 2



Heterojunction 异质结

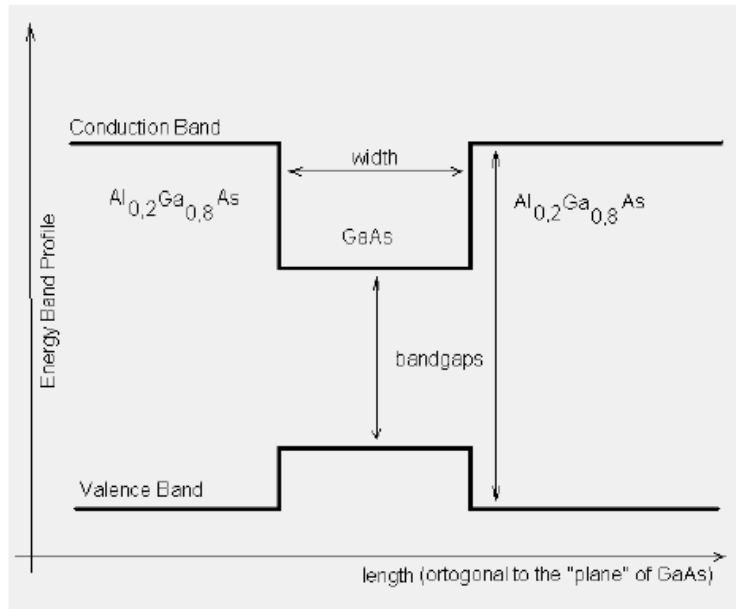


Straddling Gap

Staggered Gap

Broken Gap

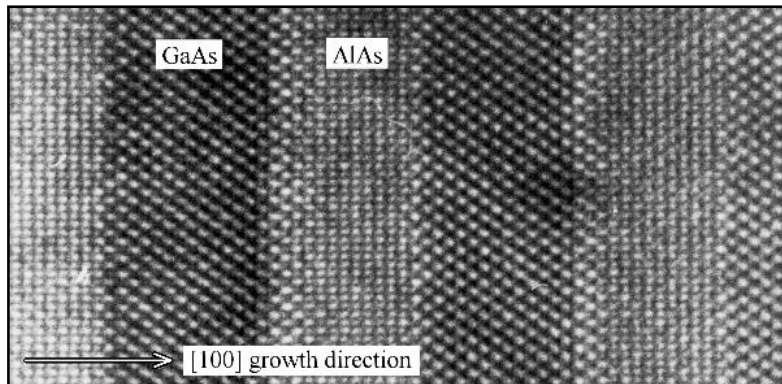
Semiconductor Heterostructures



GaAs/AlGaAs heterostructure:
Type I junction
electron and hole confinement
enhanced radiative recombination
for better LEDs and lasers



Z. I. Alferov



H. Kroemer

2000 Nobel Prize in Physics

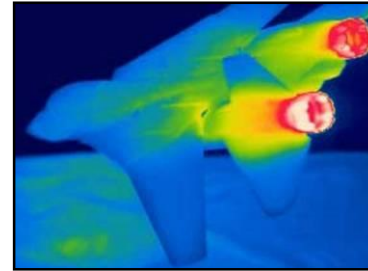
Optoelectronic Devices



LEDs



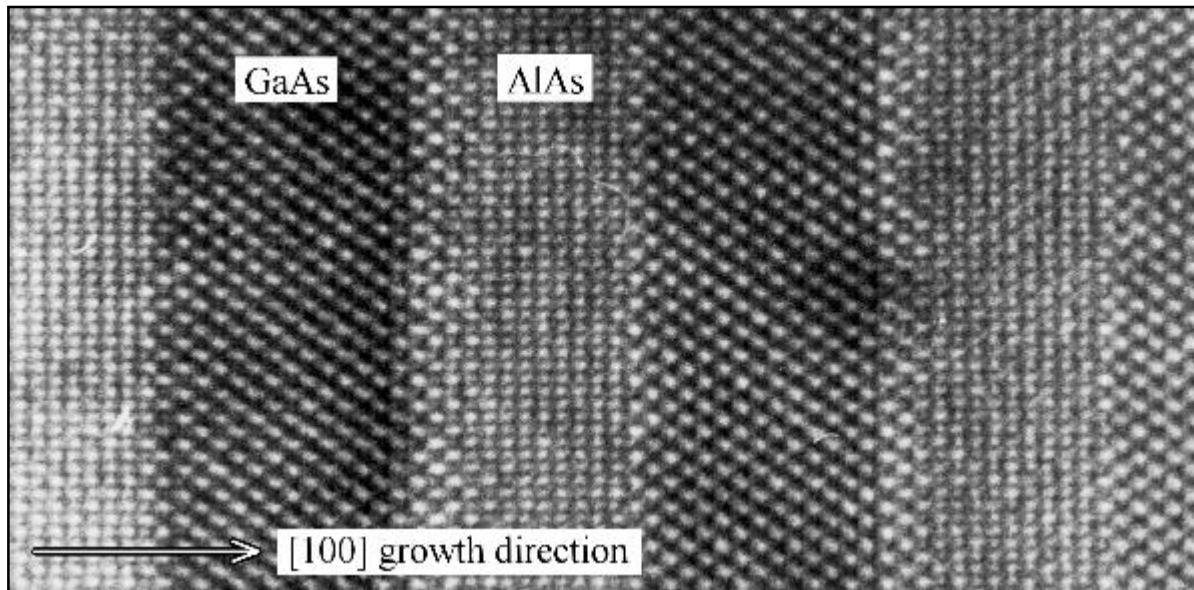
lasers



IR imaging

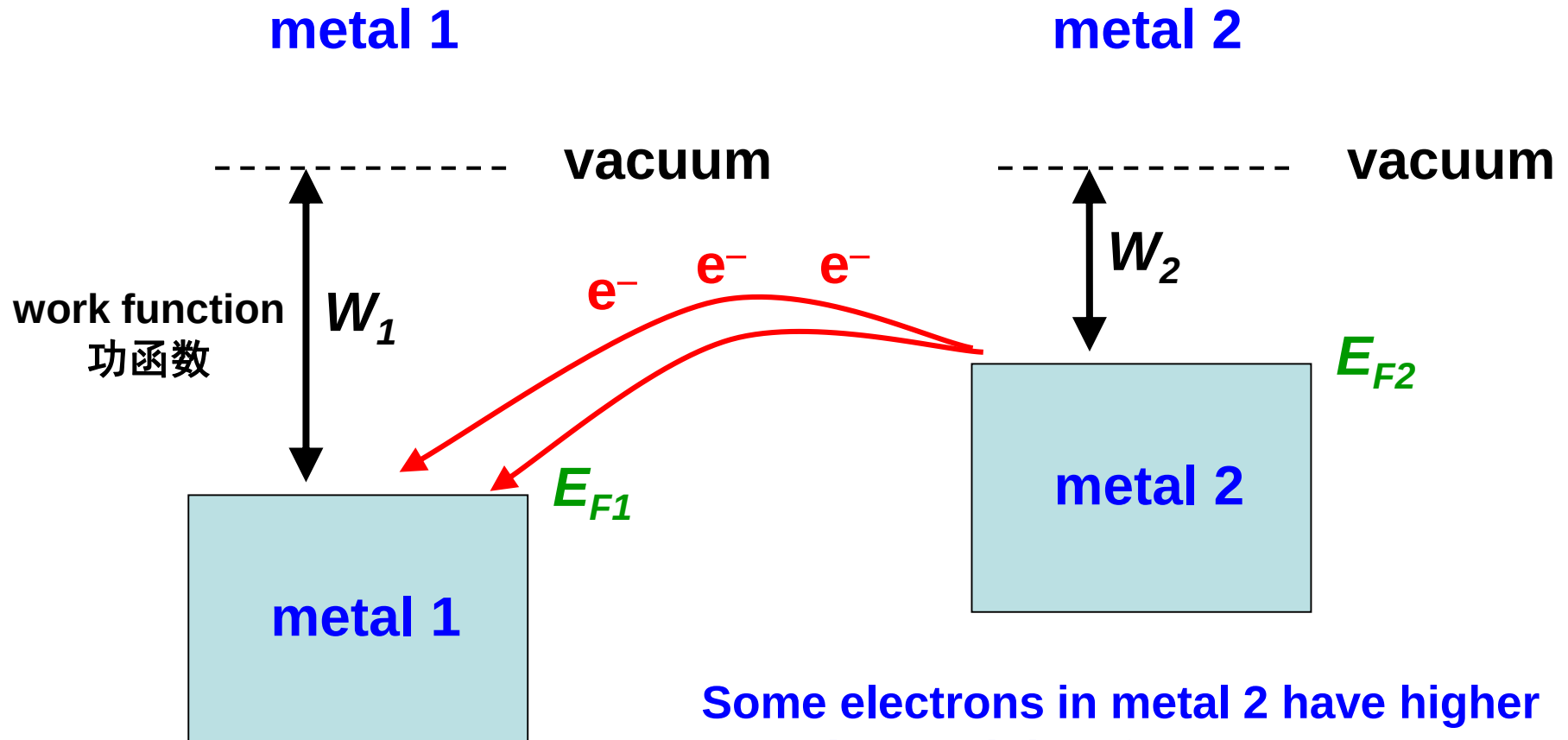


solar cells



Lattice matched GaAs/AlAs structure - perfect interface

Metal-Metal Junction

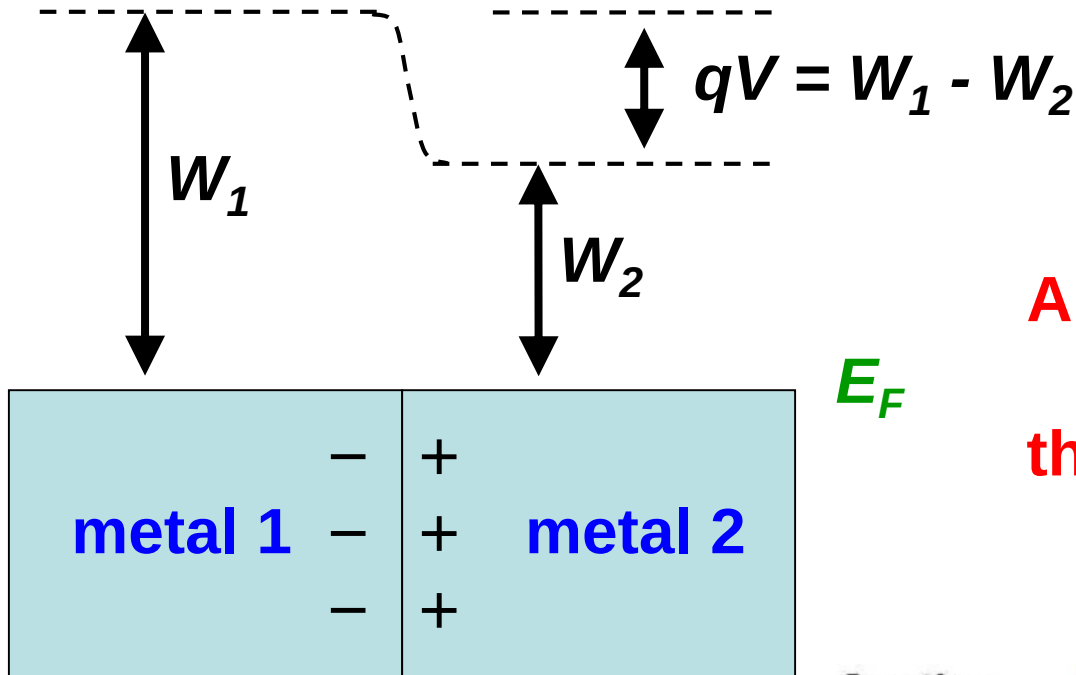


Some electrons in metal 2 have higher energies, and they flow to metal 1

metal 2 becomes more positive

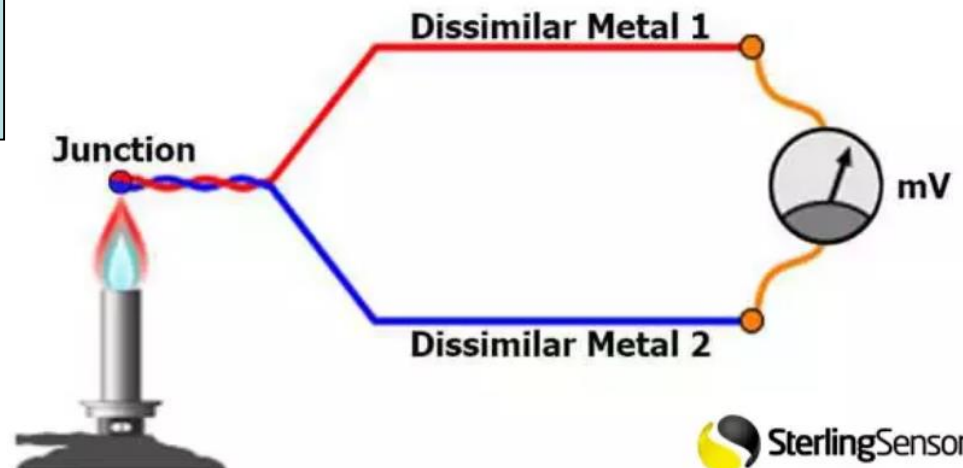
Metal-Metal Junction

contact voltage (接触势)



Application:

thermal couple 热电偶

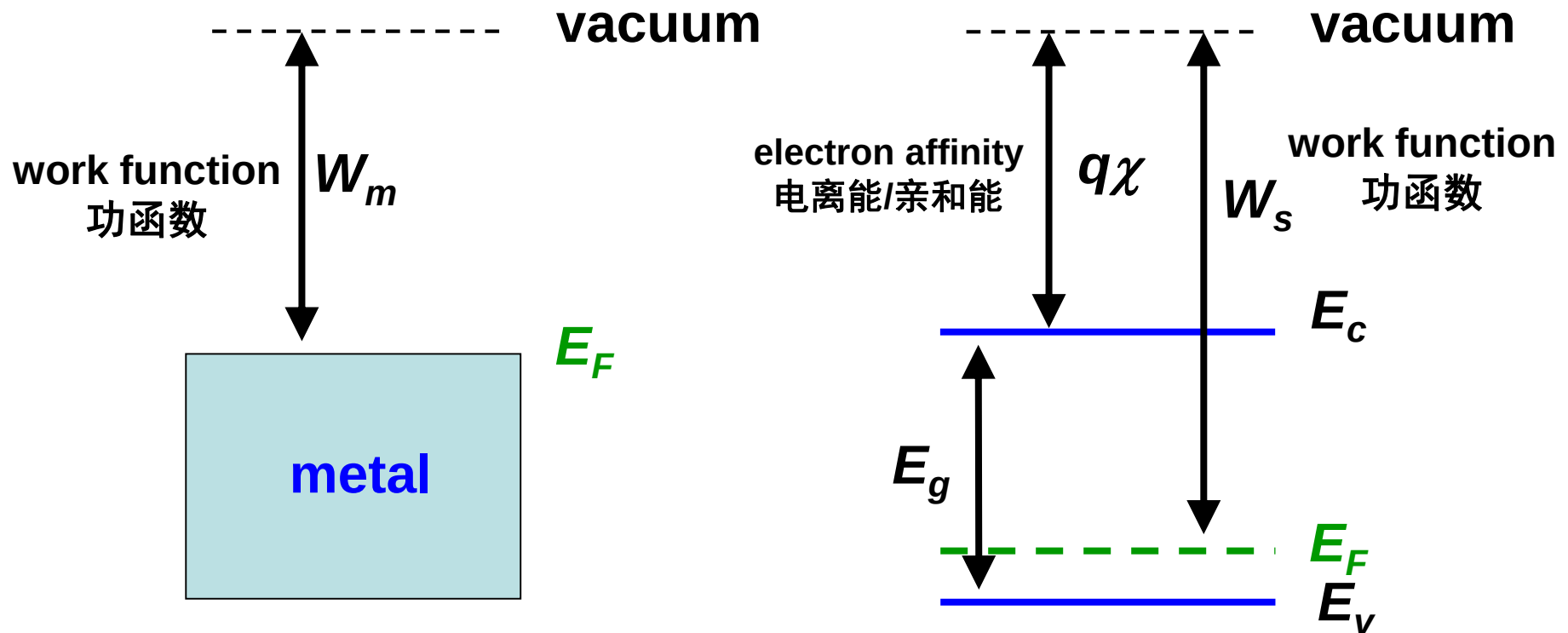


Metal-Semiconductor Junction

Case 1

metal

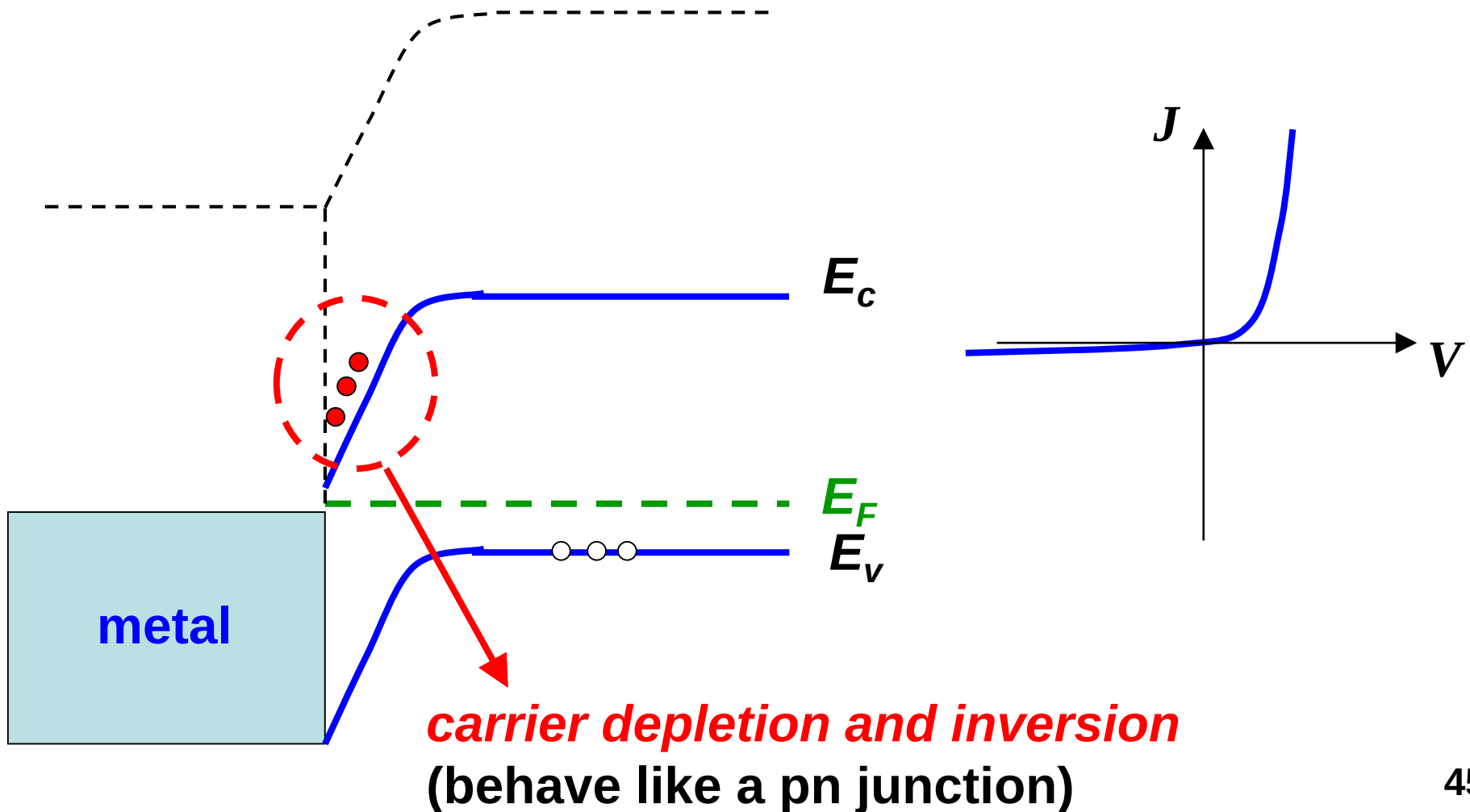
p-type semiconductor



Metal-Semiconductor Junction

Case 1

Schottky contact 肖特基接触

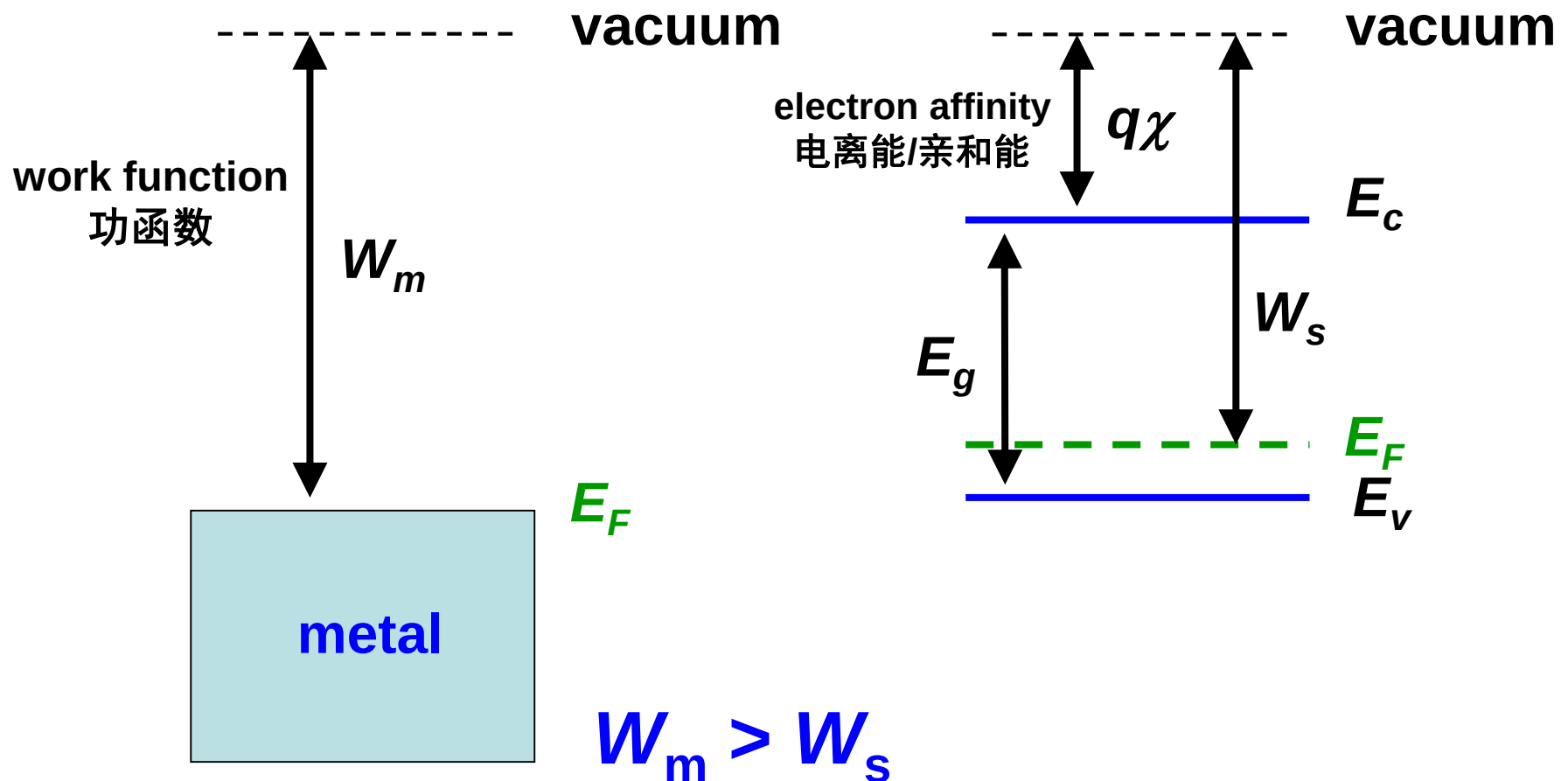


Metal-Semiconductor Junction

Case 2

metal

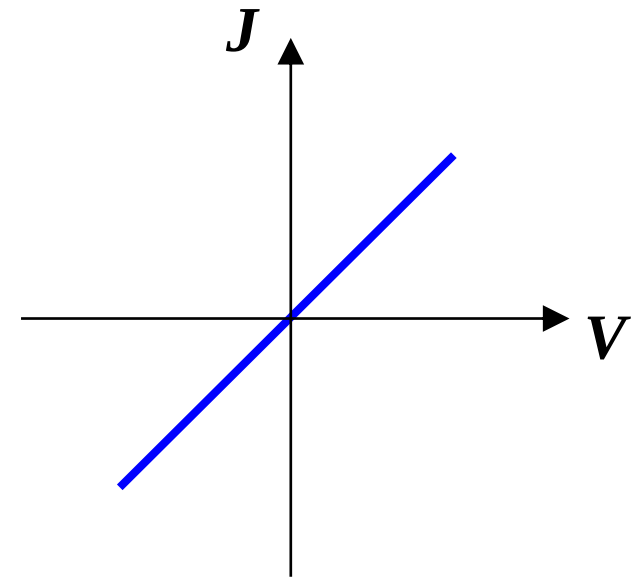
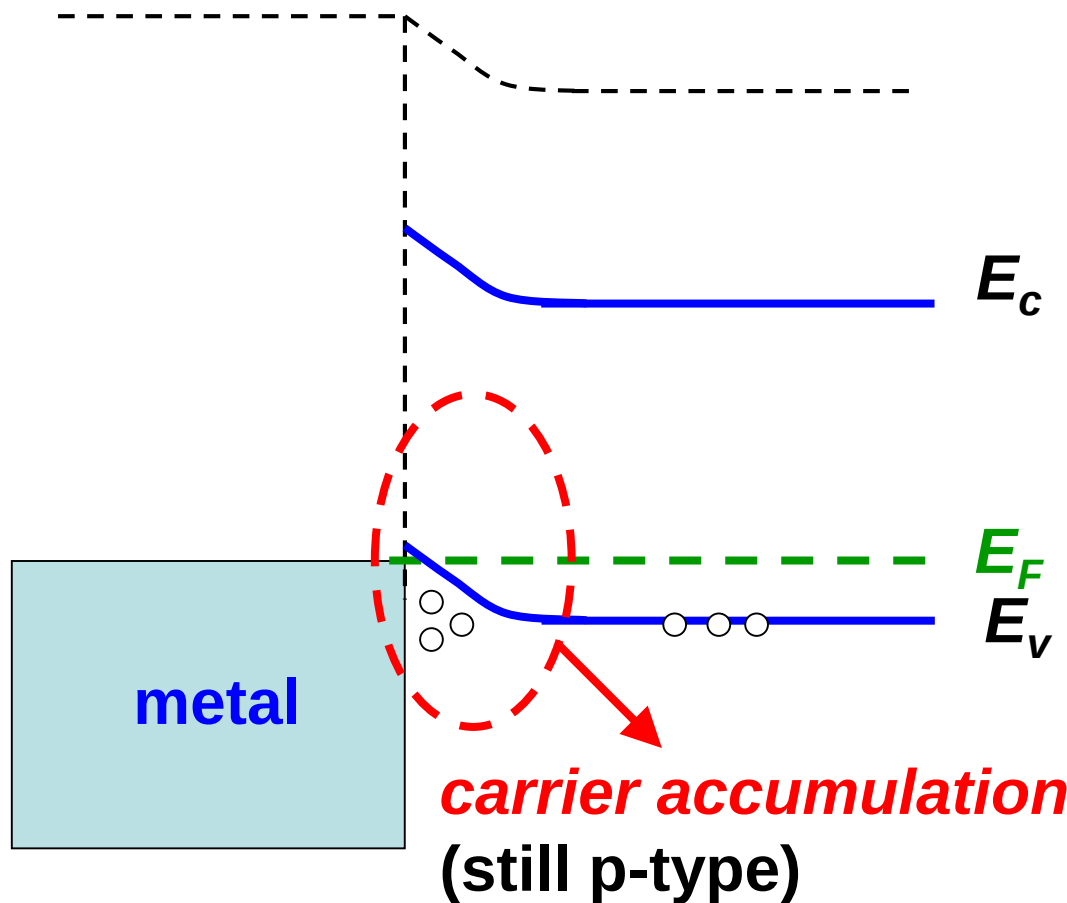
p-type semiconductor



Metal-Semiconductor Junction

Case 2

Ohmic contact 欧姆接触

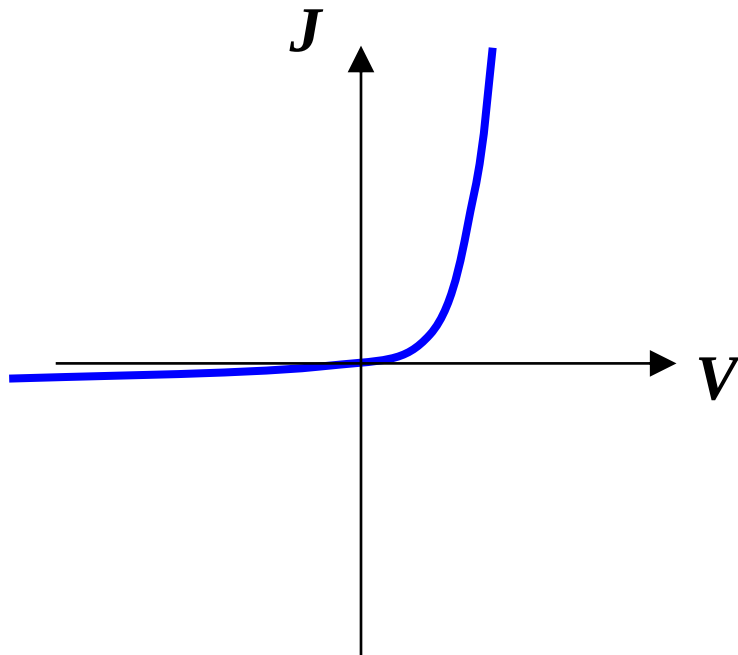


Metal-Semiconductor Junction

For n-type, when $W_m > W_s$

For p-type, when $W_m < W_s$

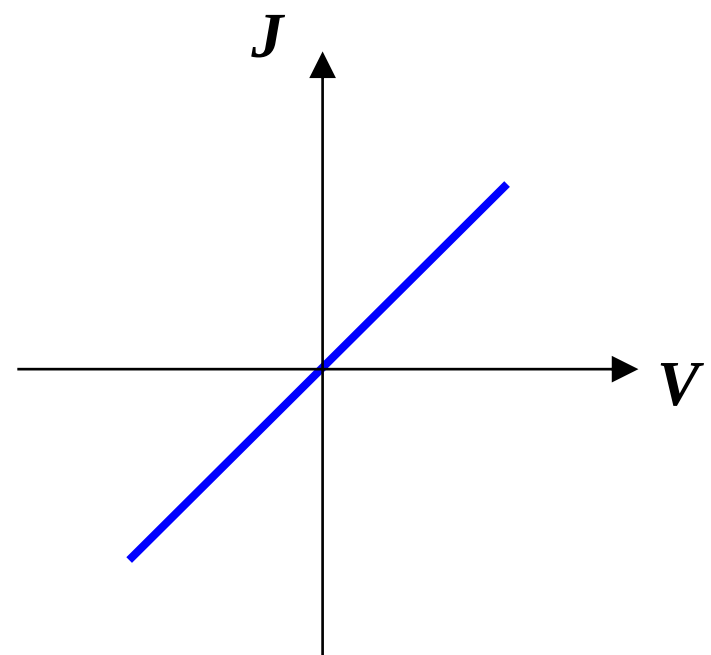
Schottky contact 肖特基接触



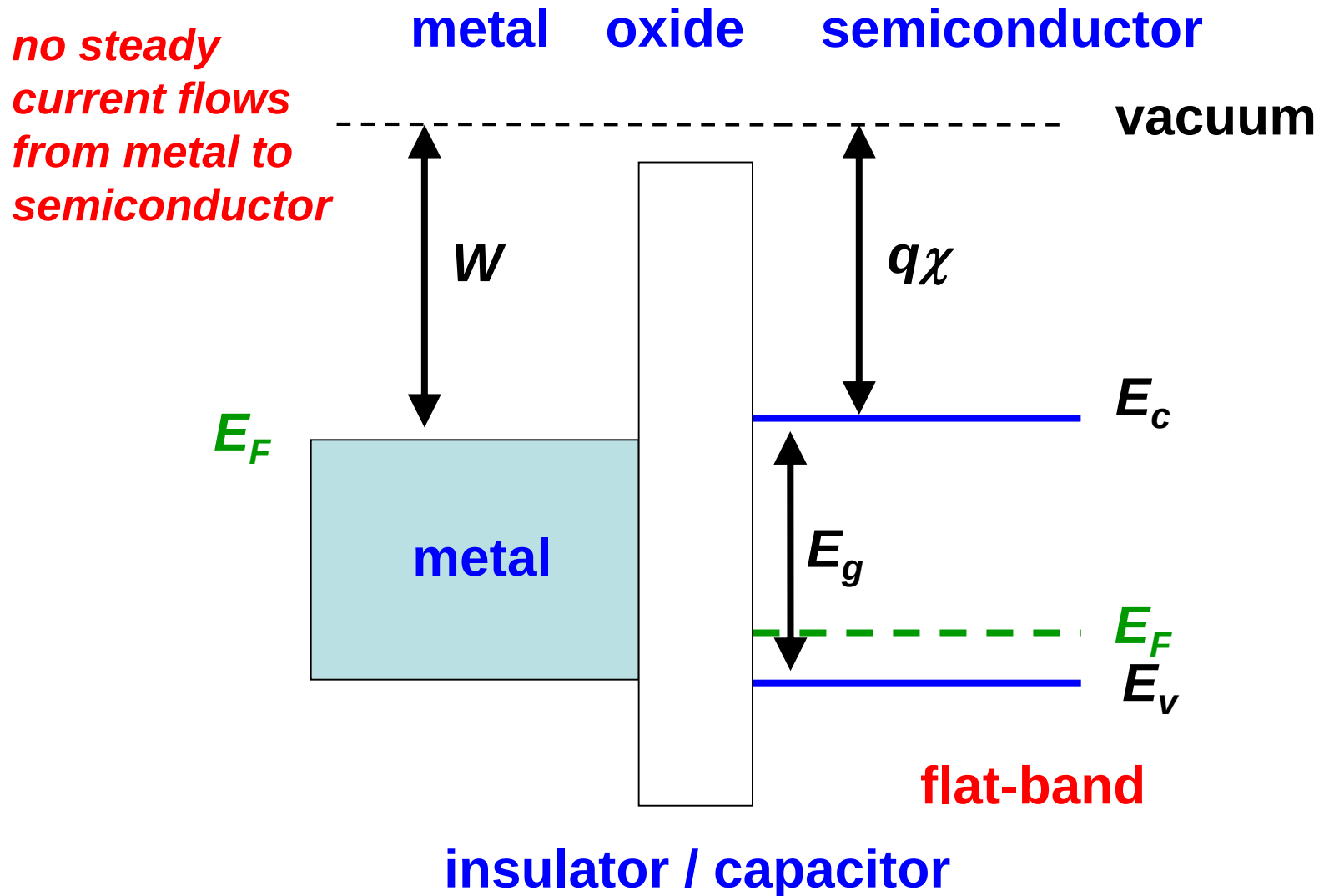
For n-type, when $W_m < W_s$

For p-type, when $W_m > W_s$

Ohmic contact 欧姆接触

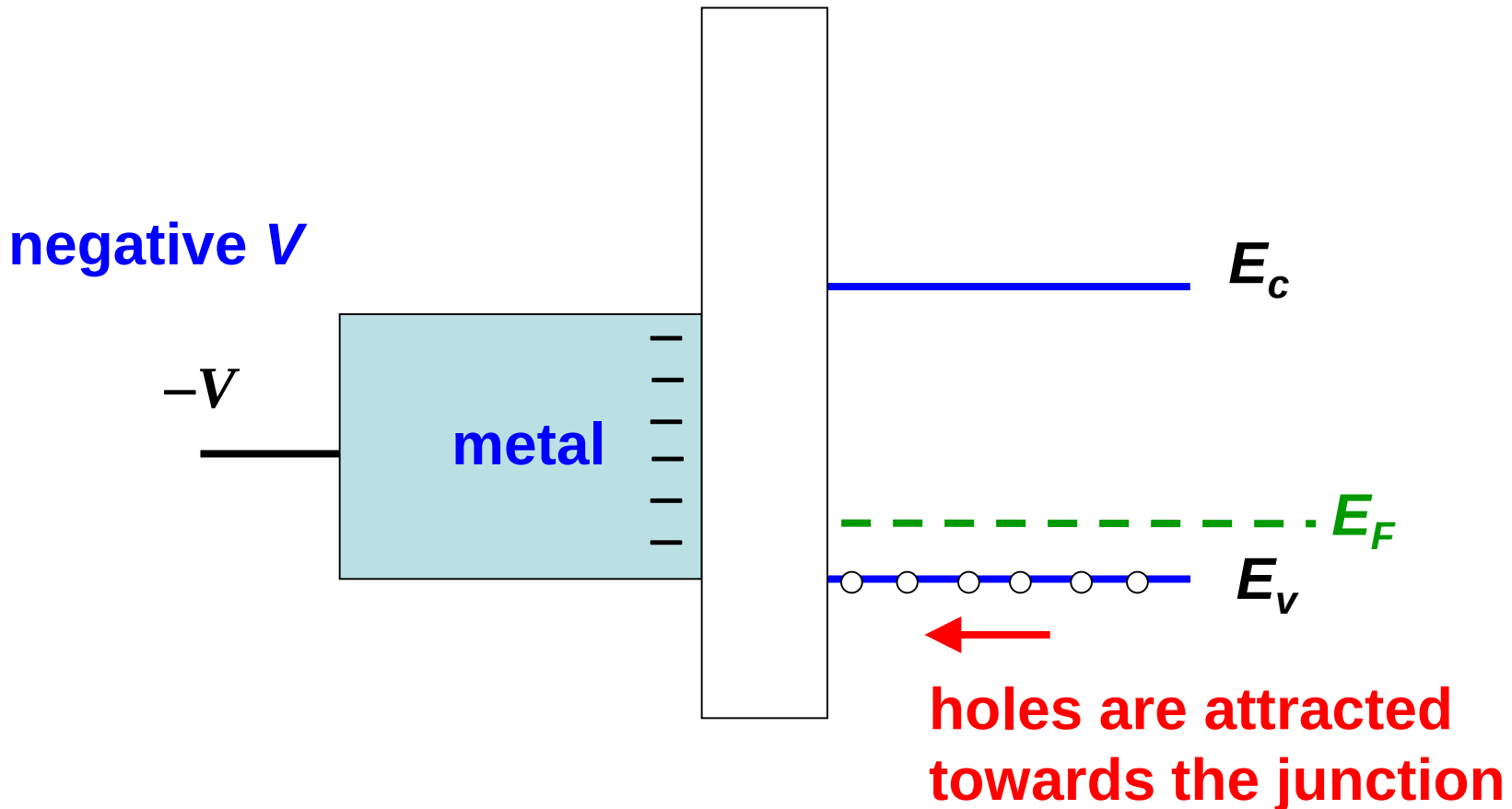


Metal-Oxide-Semiconductor Junction



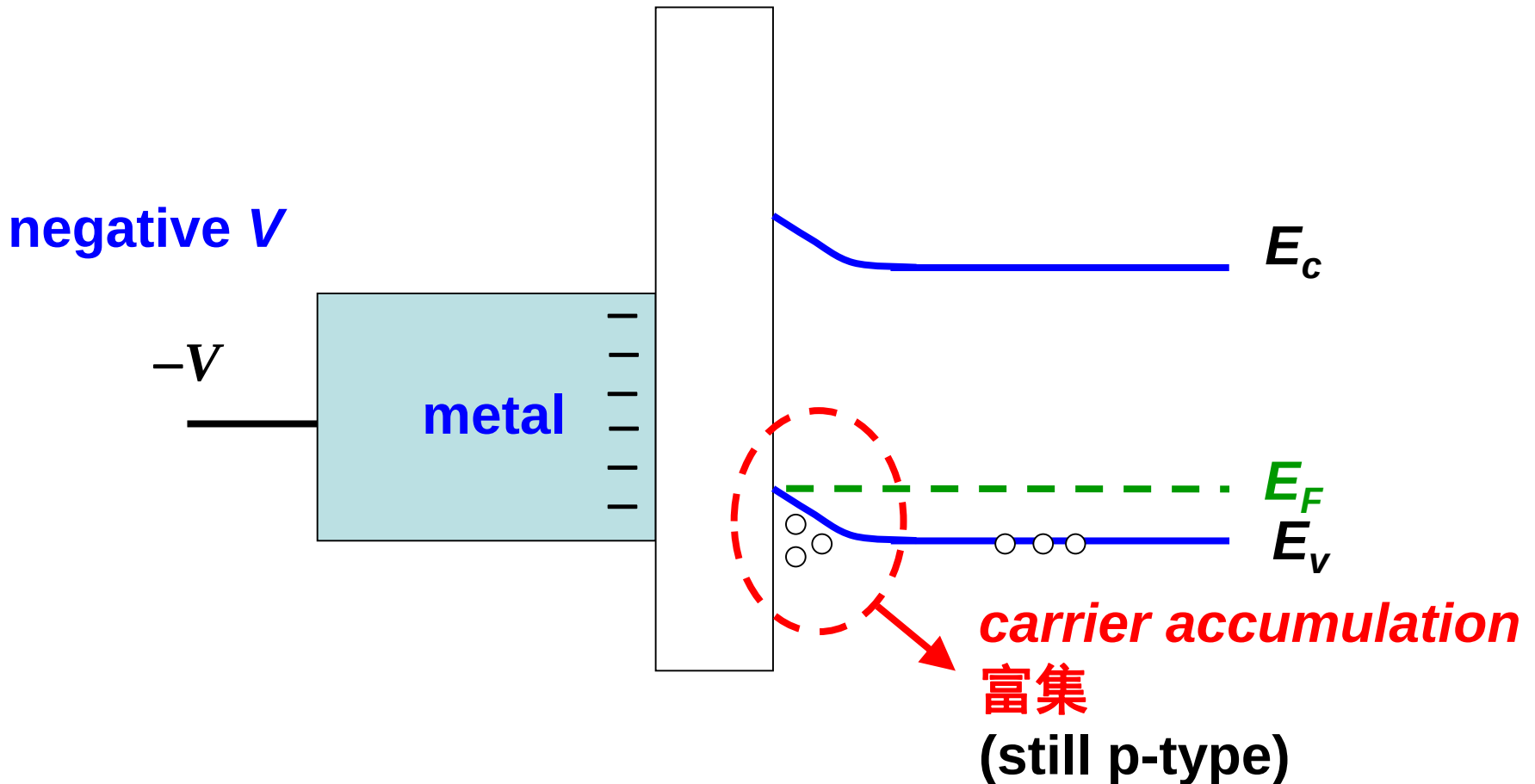
Metal-Oxide-Semiconductor Junction

metal oxide semiconductor



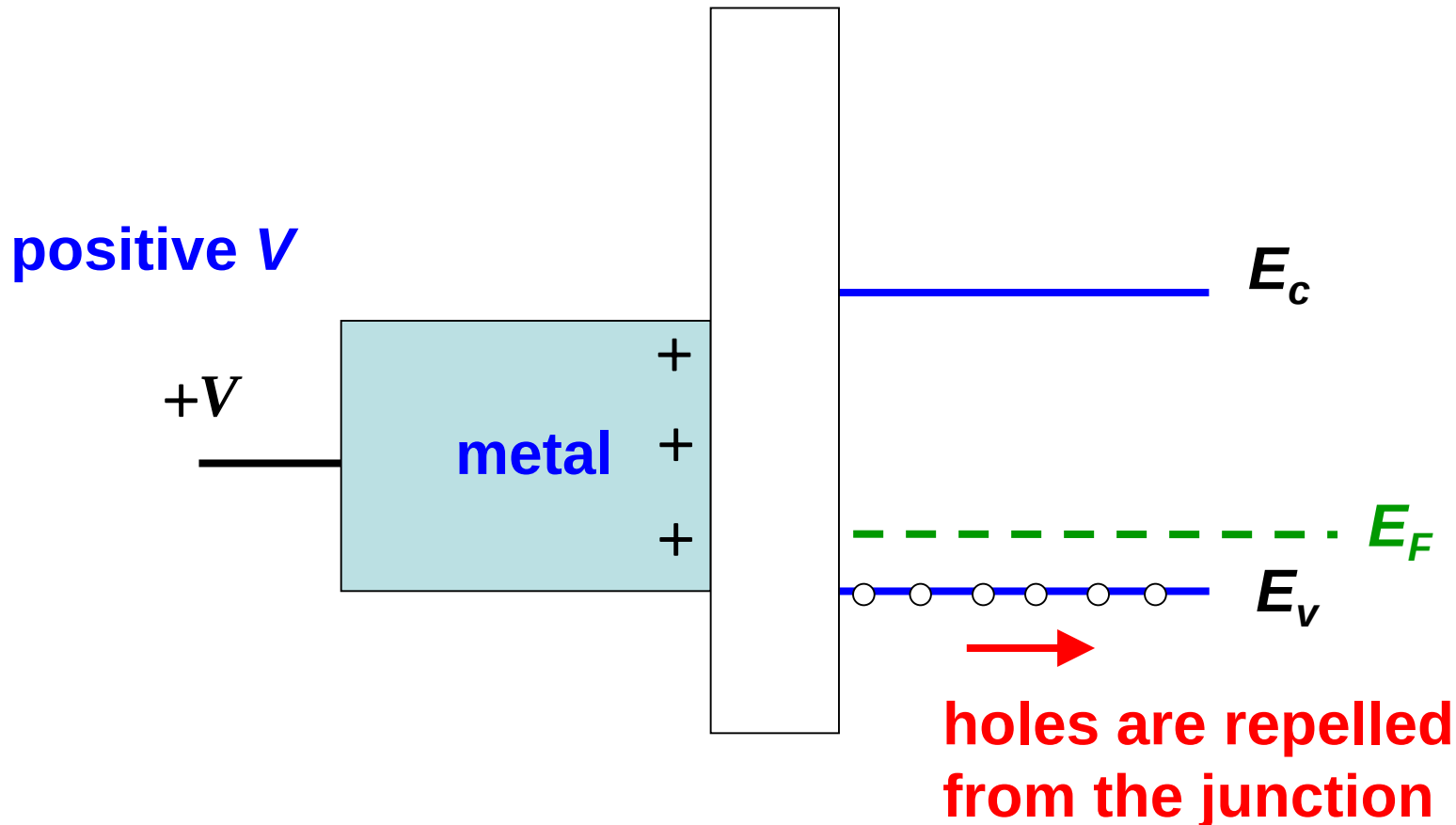
Metal-Oxide-Semiconductor Junction

metal oxide semiconductor



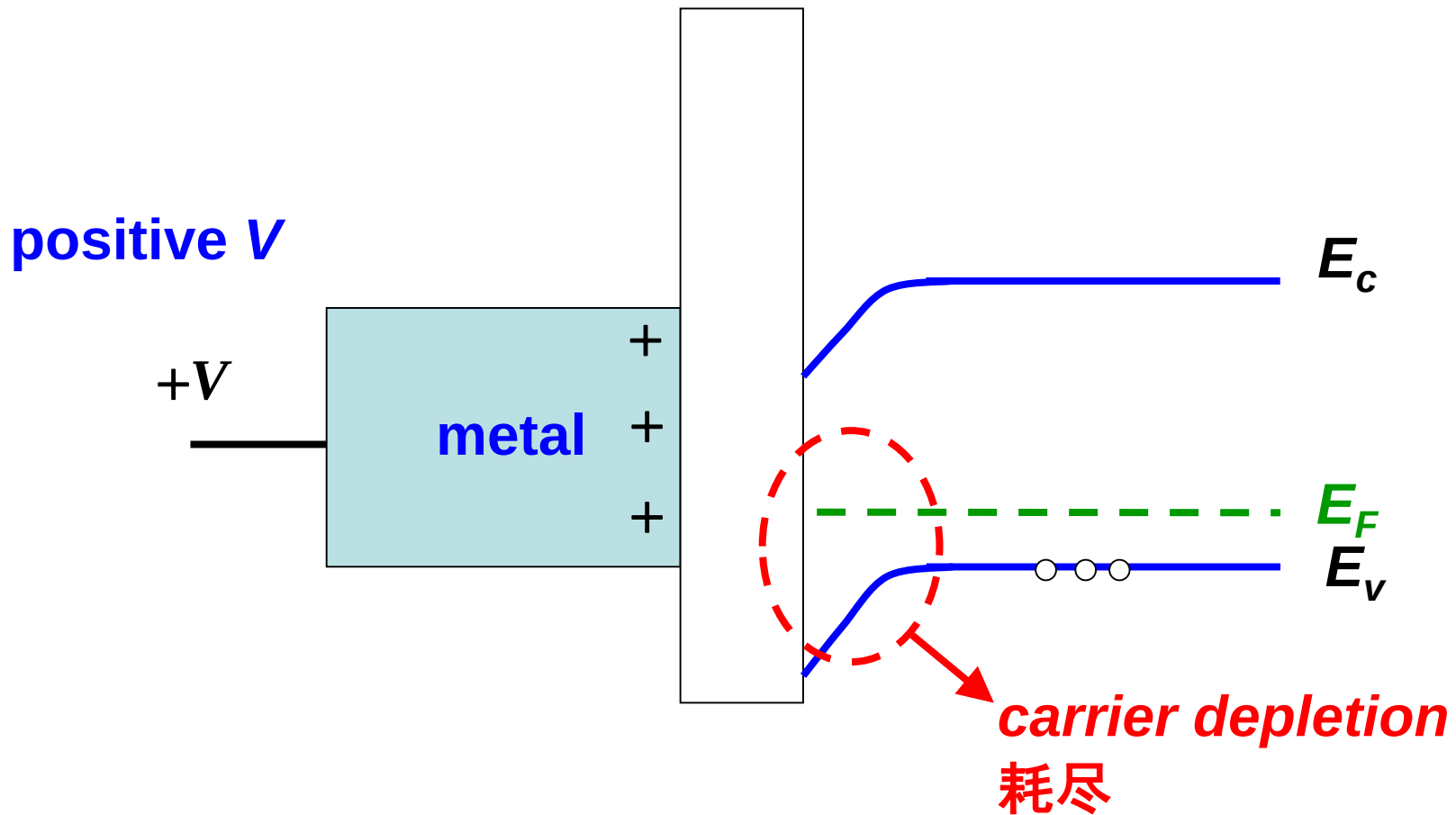
Metal-Oxide-Semiconductor Junction

metal oxide semiconductor



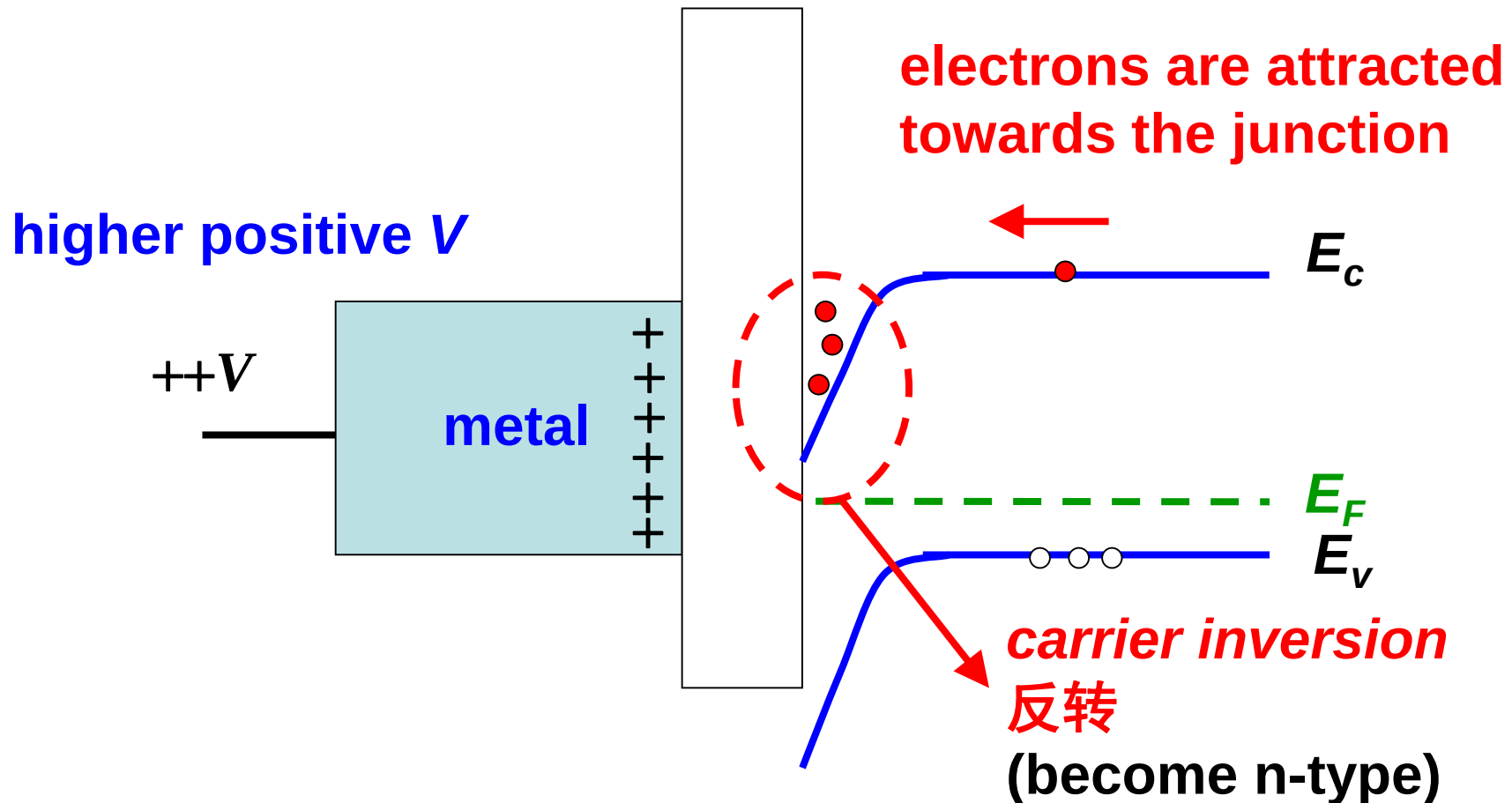
Metal-Oxide-Semiconductor Junction

metal oxide semiconductor

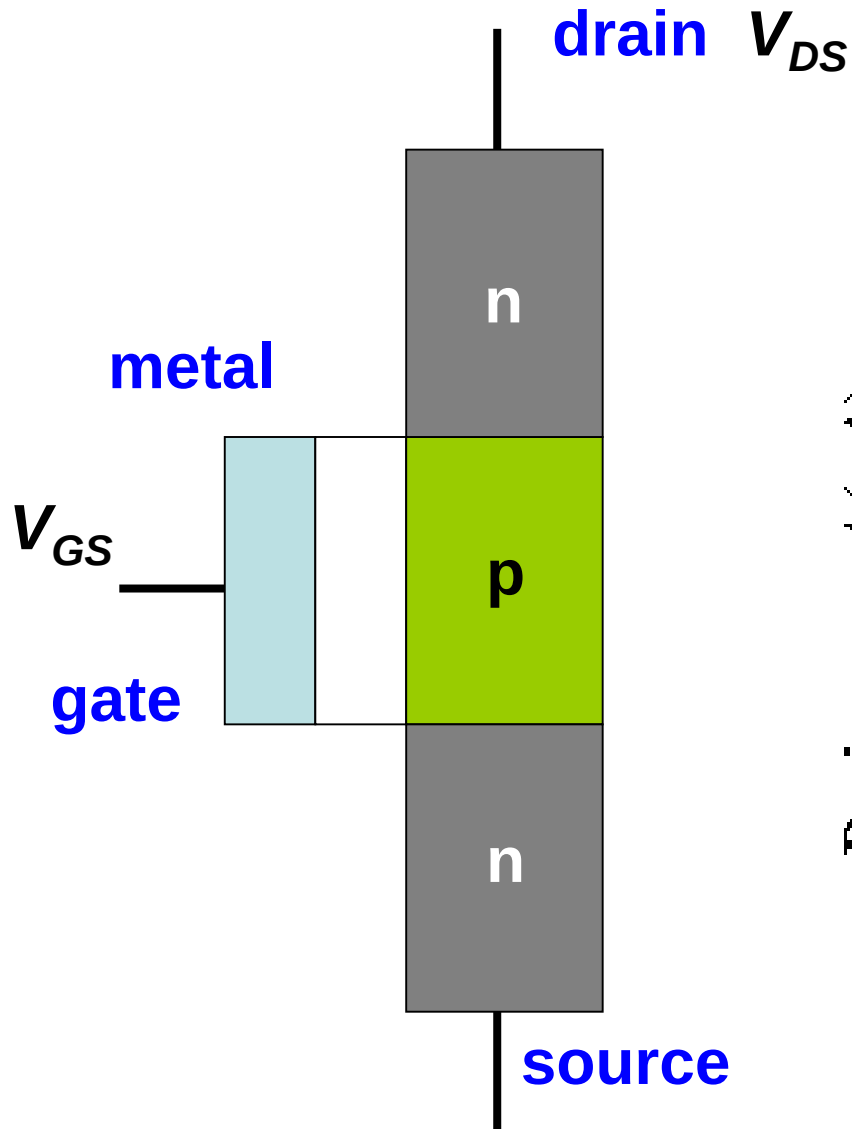


Metal-Oxide-Semiconductor Junction

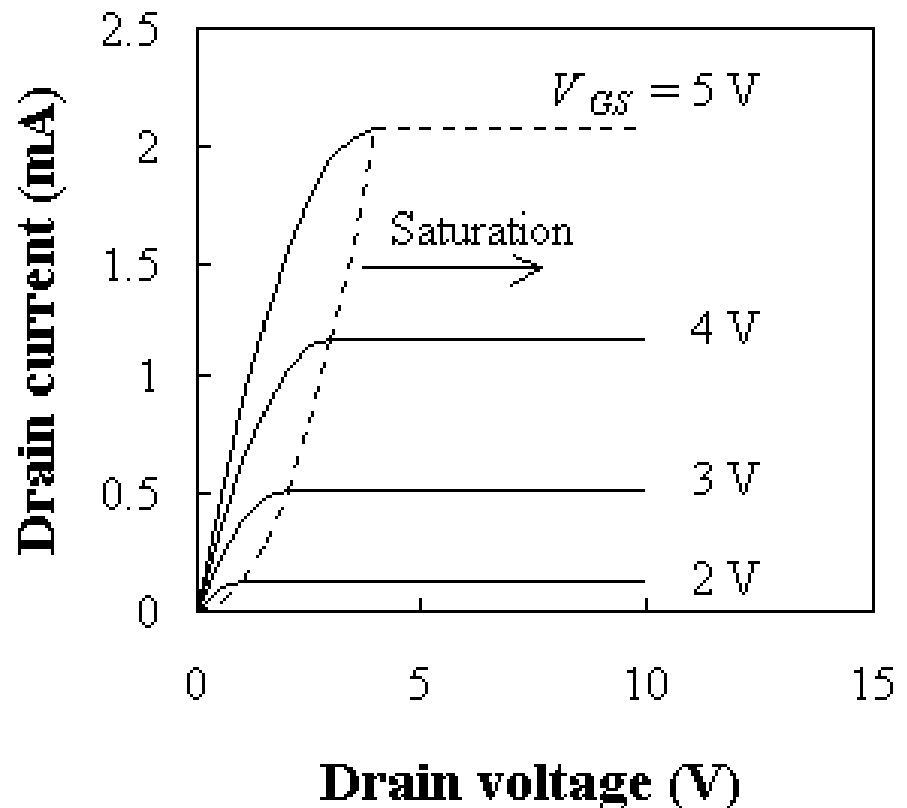
metal oxide semiconductor



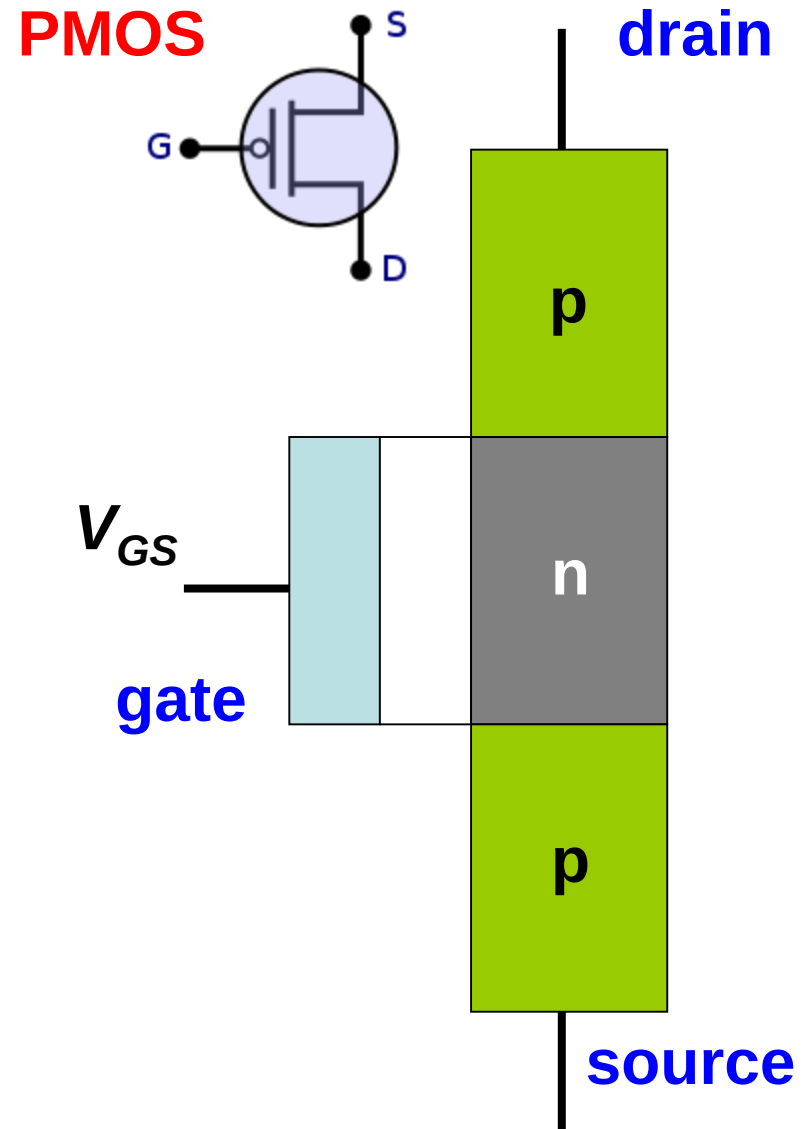
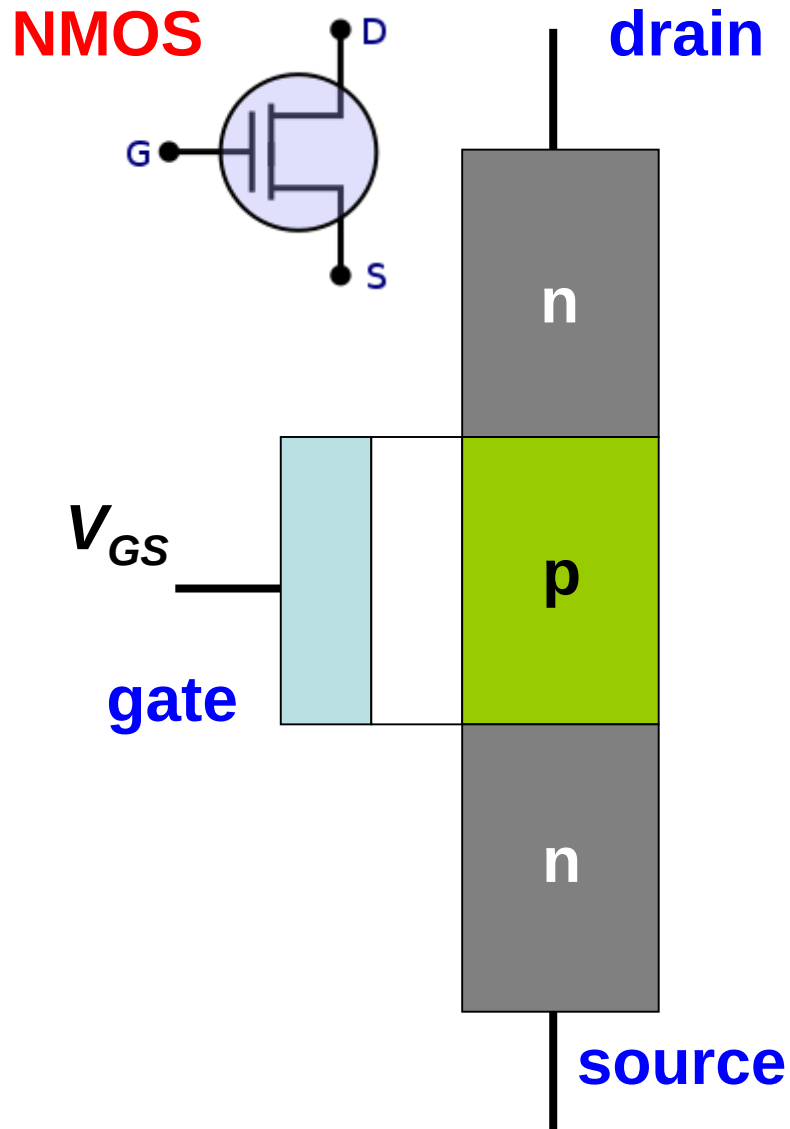
Metal-Oxide-Semiconductor Junction



**Metal-Oxide-Semiconductor
Field-Effect Transistor
MOSFET 场效应晶体管**

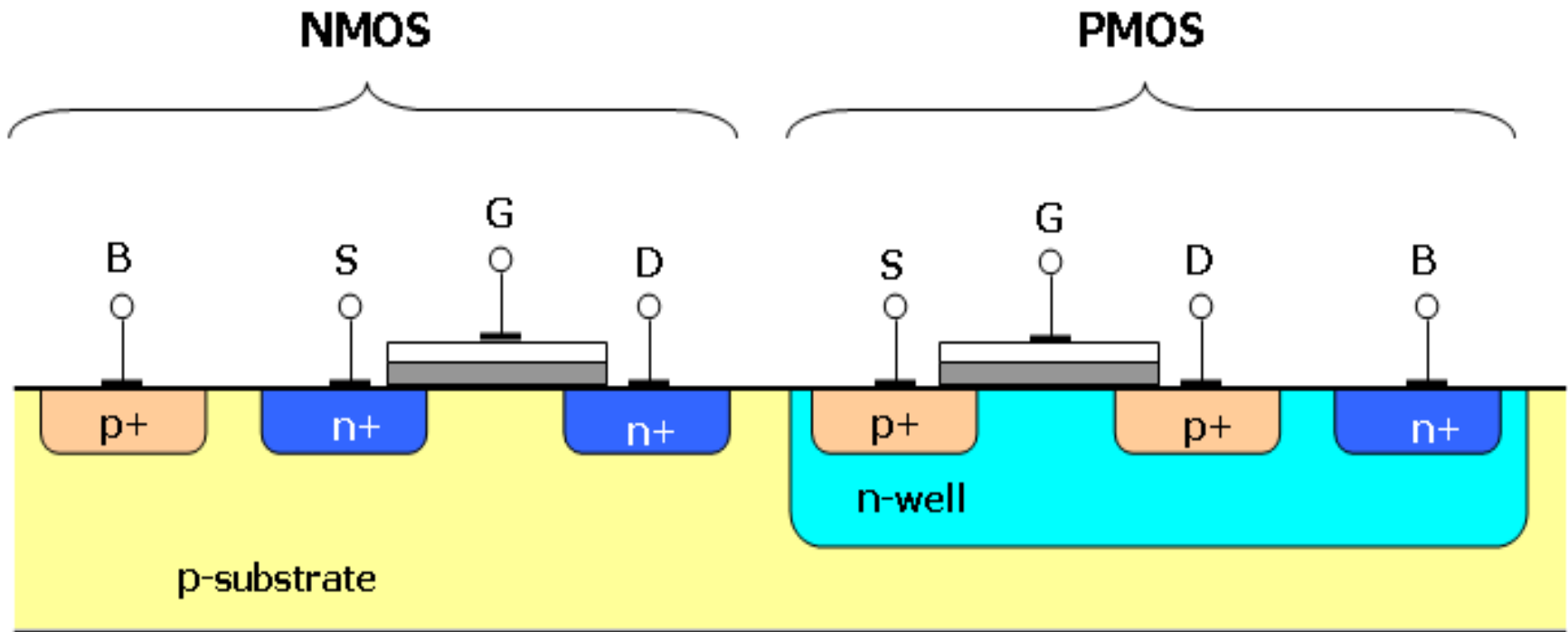


MOSFET



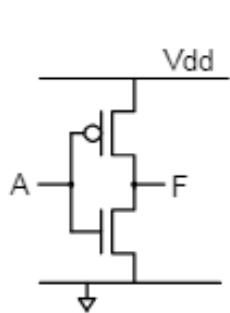
CMOS Technology

- Complementary Metal-Oxide-Semiconductor



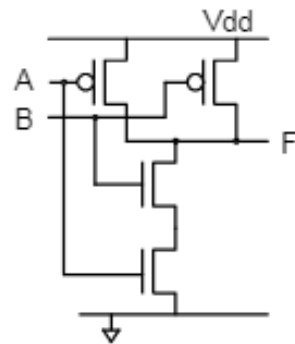
[Video MOSFET](#)

CMOS Logics



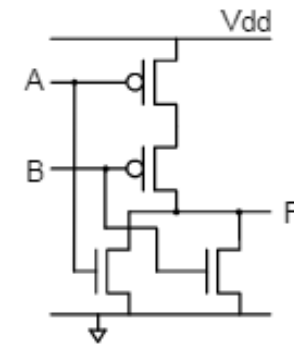
A	F
L	H
H	L

CMOS INVERTER



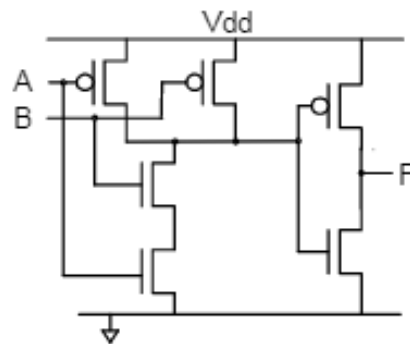
A	B	F
L	L	H
L	H	H
H	L	H
H	H	L

CMOS NAND



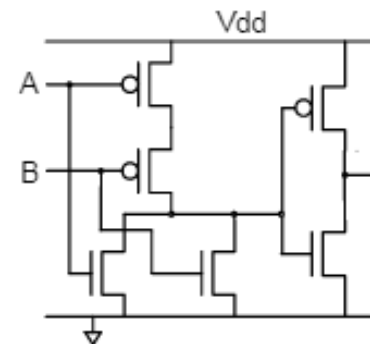
A	B	F
L	L	H
L	H	L
H	L	L
H	H	L

CMOS NOR



A	B	F
L	L	L
L	H	L
H	L	L
H	H	H

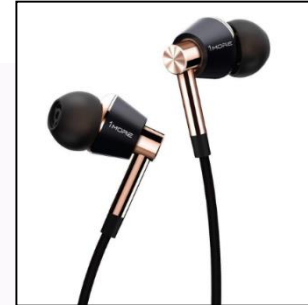
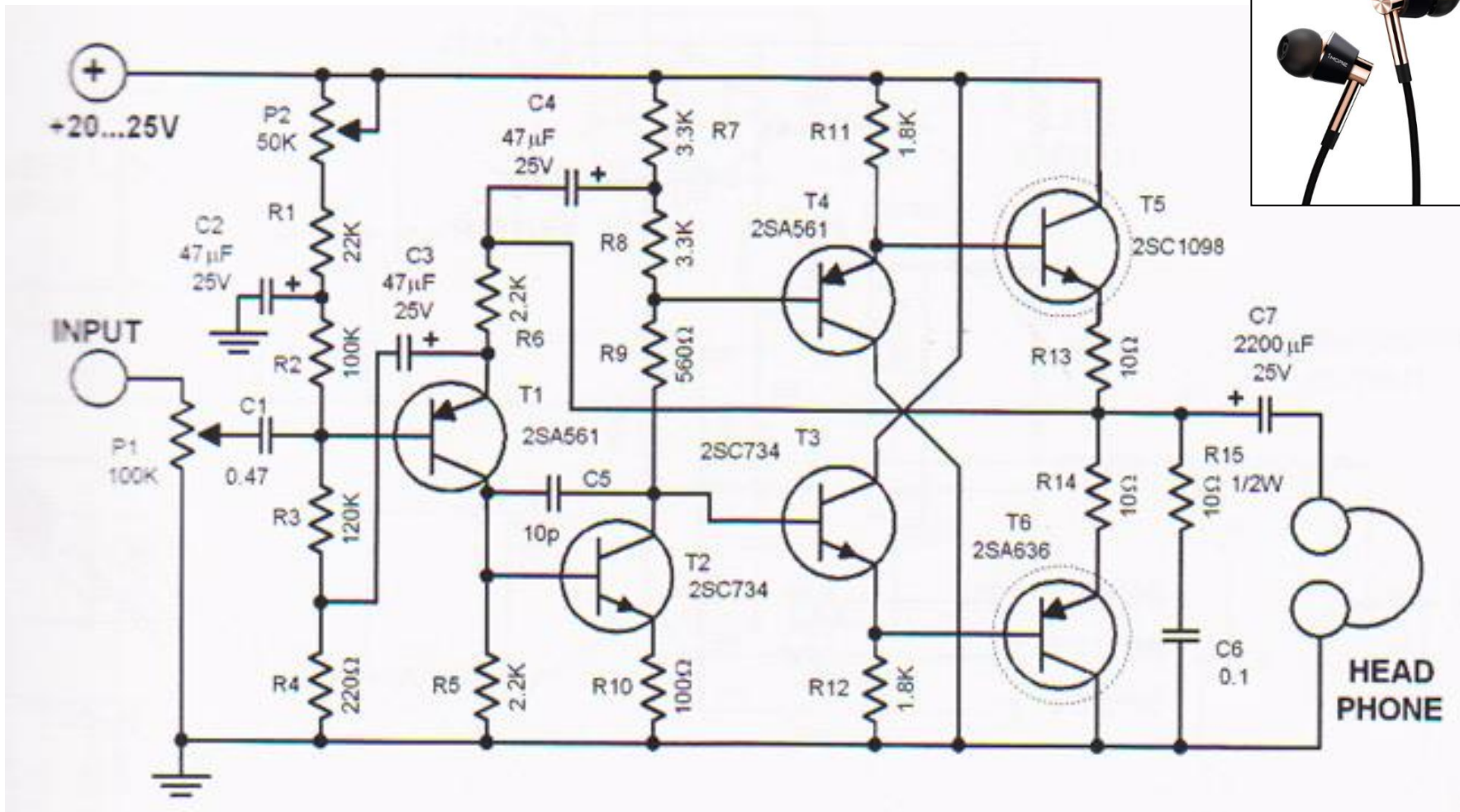
CMOS AND



A	B	F
L	L	L
L	H	H
H	L	H
H	H	H

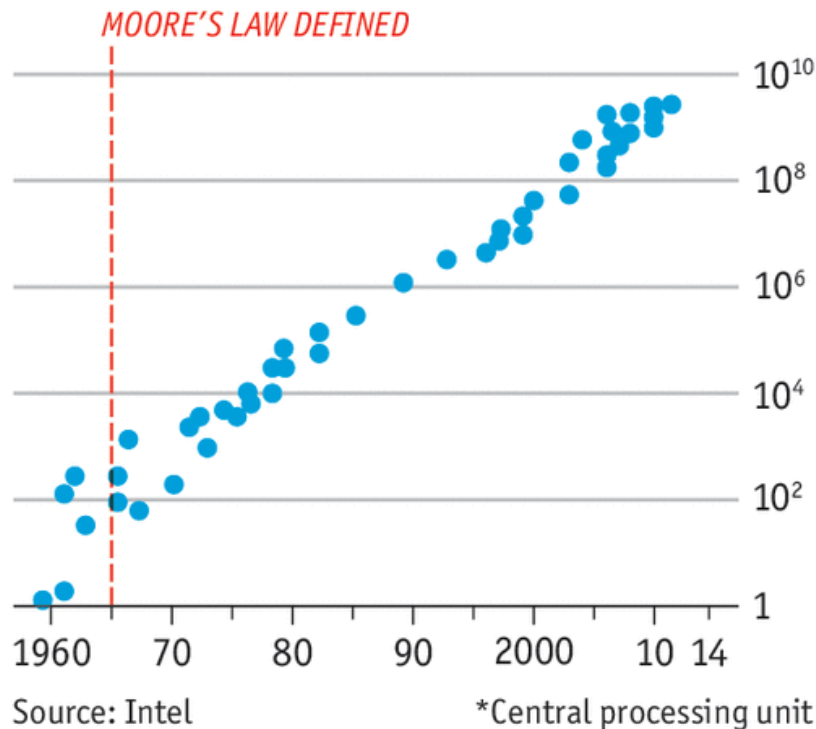
CMOS OR

CMOS Circuits



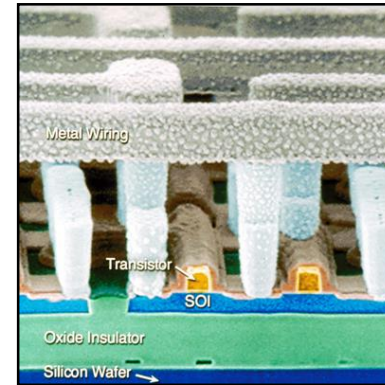
Integrated Circuits

■ Moore's law, Fairchild, 1965

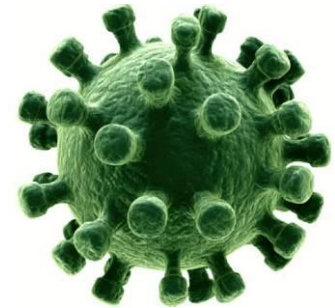


Economist.com

***Modern Electronics is a
real Nanotechnology***



← →
< 100 nm



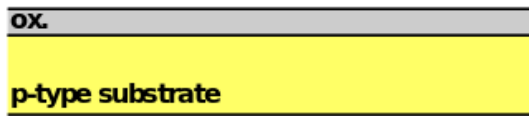
coronavirus



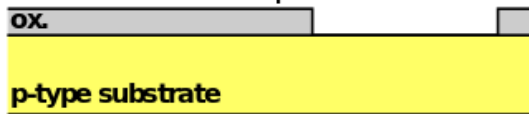
Gordon Moore
Intel i7 CPU, ~ 10⁹ transistors

CMOS Process

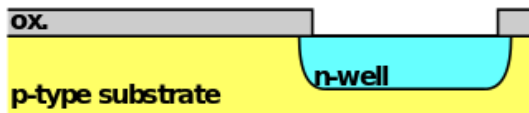
1. Grow field oxide



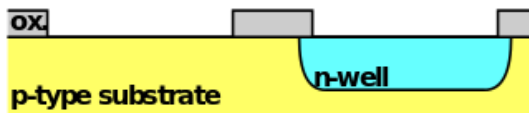
2. Etch oxide for pMOSFET



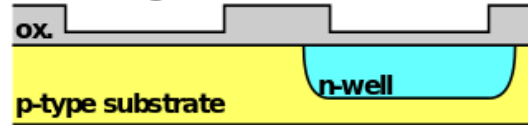
3. Diffuse n-well



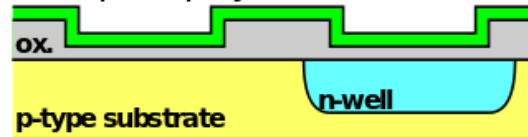
4. Etch oxide for nMOSFET



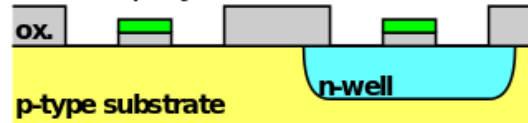
5. Grow gate oxide



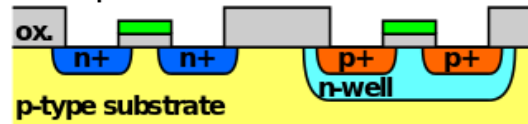
6. Deposit polysilicon



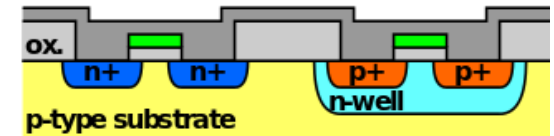
7. Etch polysilicon and oxide



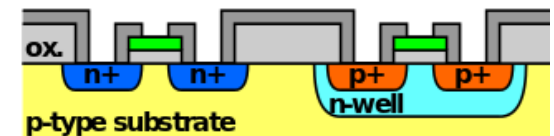
8. Implant sources and drains



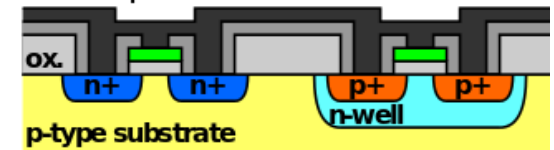
9. Grow nitride



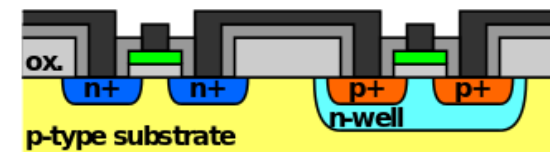
10. Etch nitride



11. Deposit metal



12. Etch metal



Course Information

- Principles of Micro- and Nanofabrication for Electronic and Photonic Devices
微纳光子材料与器件工艺原理 80230992
- Course Focus: *How to make solid-state devices?*
- Every Spring Semester
- Instructor
 - 盛兴 (SHENG Xing)
 - xingsheng@tsinghua.edu.cn

This Class

- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- **Electronic Properties (Week 4–12)**
 - Free electrons (the Drude and Sommerfeld models)
 - Electrons in a periodic potential (Bloch's Theorem)
 - The near-free electron model, the tight-binding model
 - Electronic band diagram, band gaps, effective mass
 - Metals, insulators, semiconductors
 - Devices: junctions, diodes, transistors
- Thermal Properties (Week 13)
- Optical Properties (Week 14)
- Magnetic Properties (Week 15)

Thank you for your attention